



Core unlearning: A multi-modal gradient-efficient architecture for exact and approximate model rewriting

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ABSTRACT

Machine unlearning is important for data security, user confidence, and regulatory compliance in AI systems. Despite the significant achievement, existing techniques have limited generalizability across a broad set of forgetting scenarios — feature, class, task, stream, or catastrophic forgetting, and are devoid of a theoretical base, scalability, or computational efficiency. The proposed Core Unlearning (CU) framework bypasses these limitations by integrating state-of-the-art methods like latent space loss optimization, gradient ascent-augmented updates, Adapter Partition and Aggregation (APA), and Projection-Based Residual Adjustment (PBRA) into a unified structure that supports both Exact Unlearning (EU) and Approximate Unlearning (AU). In EU, Negative Preference Optimization (NPO) is employed, a strategy that treats target data as negative samples to actively suppress their influence during unlearning by penalizing correct predictions on forgotten data. Evaluating across multi-modal datasets like CIFAR-10, CIFAR, 100, IMDB4K, CORA, FEMNIST, and MVTec AD, CU achieves improved performance in forgetting fidelity, model utility, and privacy preservation. The GA+APA+NPO achieves up to 2.3% decreased accuracy loss, with 95.2% retraining equivalence, proving high-fidelity unlearning. In AU mode, our approach gets 92.3% forgetting accuracy, 85.7% utility score, and 90.2% unlearning efficiency, enabling a scalable solution for time-critical applications. With a seamless combination of EU and AU into a single paradigm, CU enables versatile management of the precision-speed trade-off, with support for strong application-specific unlearning. The work in this paper demonstrates an early step toward useful, mathematically robust, and privacy-preserving machine unlearning. Code available at: [CoreUnlearning](https://github.com/AttiqueKhan/CoreUnlearning).

1. Introduction

Machine unlearning has emerged as a critical research discipline, driven by the increasing demand for robust data protection mechanisms in AI systems (Liu, Ye, Chen, Zheng and Lam, 2024; Wang, Tian, Zhang, & Yu, 2024). In this work, model rewriting refers to the precise, parameter-level modification of a trained model to remove the influence of specific data-such as individual

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samples, features, or entire classes-while preserving its utility on the rest of the dataset, achieved through targeted mathematical operations. As machine learning models become integral to diverse applications-from healthcare to finance-there is a growing need for such structured, programmable model updates to ensure privacy, compliance, and adaptability (Guo, Liu, Zhou, Cai, & Xiao, 2023; Li, Zhang, Zhang, & Cheng, 2025). Despite notable advancements, significant challenges persist in developing frameworks that are both mathematically rigorous and practically effective across a range of use cases (Kurmanji, Triantafillou, Hayes, & Triantafillou, 2024; Li et al., 2024, 2024; Mittal et al., 2024; Qu et al., 2024).

Among existing paradigms, Exact Unlearning (EU) and Approximate Unlearning (AU) present two dominant approaches with distinct trade-offs. EU (Bourtoule et al., 2021; Cao & Yang, 2015; Ma et al., 2022) aims to exactly erase the impact of targeted data from a trained model by performing fine-grained parameter adjustments. This method typically involves retraining or analytically computing updates that nullify the contribution of specified data points while preserving the model's performance on retained inputs. While EU offers strong theoretical guarantees regarding forgetting fidelity, it often incurs high computational costs and requires careful calibration to avoid unintended side effects such as accuracy degradation or privacy leakage (Baliga, Case, Merkle, Stephan, & Wiehagen, 2008; Chourasia & Shah, 2023; Jia et al., 2025).

In contrast, AU (Brophy & Lowd, 2021; Lin, Zhang, Chen, Chen, & Susilo, 2023; Nguyen, Low, & Jaillet, 2020) provides a more computationally efficient alternative by approximating the unlearning process through gradient manipulation, weight pruning, or projection-based techniques. AU sacrifices some degree of precision in exchange for scalability and speed, making it suitable for large-scale or time-sensitive applications. However, this efficiency comes at the cost of potential residual data influence and imperfect forgetting, particularly when complex dependencies exist between training samples (Liu et al., 2024; Wichert & Sikdar, 2024; Zhao, Huang, Li, Chen, & Wang, 2025).

To unify these complementary strategies, we propose CU — a comprehensive framework that integrates EU and AU into a single, modular architecture. CU enables seamless transitions between exact and AU based on application-specific requirements, such as the sensitivity of removed data, available computational resources, and tolerance for approximation error. By leveraging forward-looking numerical techniques – including latent space loss optimization, gradient ascent-enhanced updates, and adaptive pruning – CU ensures both theoretical soundness and practical effectiveness across diverse scenarios.

The CU framework introduces novel formulations of unlearning loss functions, such as $L_{forget} = -\sum_{i=1}^N \log(1 - P(y_i|x_i, W)) + \tau \cdot \|\nabla_W P(y_i|x_i, W)\|_2^2 + \kappa \cdot \|H^{-1} \cdot \nabla_W P(y_i|x_i, W)\|^2$, which balance the reversal of learned patterns with model stability and generalization preservation. Additionally, CU incorporates modules like Adapter Partition and Aggregation (APA) which is a complementary and more advanced approach that leverages the principles of parameter efficient fine-tuning (Chen et al., 2022; Houlsby et al., 2019; Jie & Deng, 2022; Zhang, Zhou, & Liu, 2024) and Projection-Based Residual Adjustment (PBRA) to enhance modularity and computational efficiency. This study presents the foundational design of CU, particularization its notional principles, algorithmic components, and key advantages over existing unlearning methods. In particular, we exhibit how CU effectively bridges the gap between precise, costly EU and scalable, approximate AU, offering a flexible and robust solution for modern machine unlearning needs.

1.1. Research gap

Despite significant progress in machine unlearning, fundamental challenges persist in developing strong, generalizable, and theoretically sound techniques-particularly in the domains of CU, EU, and AU. Most existing methods are confined to narrow settings, such as sample-level data removal (Cooper et al., 2024; Hutson, Whitney, & Conrad, 2025; Liu et al., 2025), and fail to address the broader needs of modern AI systems. To guide future research, we identify the following key research gaps:

1. **Lack of Generalization Across Removal Tasks:** Current unlearning frameworks lack the ability to handle diverse forgetting scenarios beyond individual data points. A critical need exists for a unified framework capable of supporting feature-level, class-level, task-level, and stream-based forgetting across different model architectures. Mathematically, for a model $M(\theta)$ trained on $D = \{(x_i, y_i)\}_{i=1}^N$, unlearning aims to compute new parameters θ' such that:

$$\begin{aligned} P(M(\theta')(x) = y) &\approx P(M(\theta)(x) = y), \quad \forall x \notin D_{forget} \\ P(M(\theta')(x) = y) &\approx P_R(x, y), \quad \forall x \in D_{forget} \end{aligned} \quad (1)$$

In this formulation, $M(\theta)$ denotes the original model with parameters θ , trained on the full dataset D . The goal is to obtain updated parameters θ' such that the new model $M(\theta')$ behaves identically to $M(\theta)$ on all data *not* in the forget set D_{forget} . Here, $D_{forget} \subset D$ is the subset of data to be unlearned, which can correspond to individual samples, features, classes, or tasks. The first condition ensures utility preservation on retained data D_{retain} . The second condition requires that predictions on D_{forget} approximate those of a randomly initialized model $\mathcal{R}$, where $P_R(x, y)$ represents the prediction distribution of $\mathcal{R}$. While this objective is well-defined, no existing framework consistently achieves it across modalities.

2. **Absence of Standardized Benchmarking and Evaluation Protocols:** The lack of a standardized, principled evaluation protocol hinders fair comparison of unlearning methods. Composite metrics of the form $B(M) = \alpha \cdot \text{Acc}(M) + \beta \cdot P(M) + \eta \cdot C(M)$ have been informally adopted but suffer from lack of normalization, leading to dimensional inconsistency, absence of theoretical grounding, with arbitrary weight selection, non-uniform definitions across studies, preventing reproducibility, and poor discriminative power, where high scores mask significant residual data influence. As demonstrated in our experiments, such metrics fail to distinguish between superficial and effective unlearning on SOTA methods like SISA (Bourtoule et al.,

2021), GraphEraser (Chen et al., 2022), and FedEraser (Liu, Ma, Yang, Wang, & Liu, 2021). To address this, we propose the Core Unlearning Benchmark Score (CUBS):

$$B(\mathcal{M}) = \alpha \cdot \widetilde{\text{Acc}}(\mathcal{M}) + \beta_{\text{cubs}} \cdot \widetilde{\mathcal{P}}(\mathcal{M}) + \eta \cdot \widetilde{\mathcal{C}}(\mathcal{M}), \quad (2)$$

where $\widetilde{\text{Acc}}(\mathcal{M})$, $\widetilde{\mathcal{P}}(\mathcal{M})$, and $\widetilde{\mathcal{C}}(\mathcal{M})$ are min-max normalized values of accuracy retention, privacy leakage reduction, and computational efficiency, respectively. The weights α , β_{cubs} , and η are application-specific and satisfy $\alpha + \beta_{\text{cubs}} + \eta = 1$. CUBS is derived from a constrained optimization framework, ensuring dimensional consistency and enabling fair, reproducible evaluation.

3. **Inadequate Theoretical Guarantees and Privacy Quantification:** Many unlearning methods lack formal theoretical guarantees on convergence, correctness, or stability. Furthermore, quantifying information leakage under differential privacy remains underexplored $\epsilon = \log \left(\sup_{S \subseteq \text{Range}(M)} \frac{\Pr[M(D) \in S]}{\Pr[M(D') \in S]} \right)$. Here, ϵ quantifies the privacy budget in differential privacy (DP), where D and D' are adjacent datasets differing by at most one sample (i.e., $D' = D \setminus \{z\}$ for $z \in D$), and S is any measurable subset of the model's output space. The expression measures the maximum log-ratio of the probability that the model outputs fall within S when trained on D versus D' . A smaller ϵ indicates stronger privacy. This limits the empirical analysis of forgetting fidelity and prevents rigorous compliance with privacy regulations.
4. **Poor Adaptability to Dynamic and Streaming Environments:** Existing methods assume stationary data distributions and are ill-suited for streaming data or concept drift scenarios. Let Δ_t denote the drift at time t . The forgetting error evolves as:

$$\mathcal{E}_t = \mathcal{E}_{t-1} + \gamma \cdot \|\nabla_{\theta} L_{\text{forget}}^{(t)} - \nabla_{\theta} L_{\text{forget}}^{(t-1)}\|_2 \quad (3)$$

In this dynamic setting, $\mathcal{E}_t$ represents the cumulative forgetting error at time t , which accumulates based on the change in the gradient of the unlearning loss L_{forget} between consecutive time steps. The term $\nabla_{\theta} L_{\text{forget}}^{(t)}$ is the gradient of the forgetting objective at time t , and γ is a scaling factor. This highlights the need for adaptive regularization and online unlearning procedures that can dynamically adjust to changing data distributions.

5. **Limited Support for Fine-Grained Forgetting in Multi-Task Models:** Beyond sample-level forgetting, feature- and task-level forgetting are increasingly required in high-stakes domains. For a multi-task model $M(\theta) = \{T_k(\theta)\}_{k=1}^K$, erasing task T_j without disrupting others requires solving a constrained optimization problem:

$$\min_{\theta'} \sum_{k \neq j} \mathcal{L}(T_k(\theta'), D_k) + \lambda \cdot \mathcal{L}_{\text{forget}}(T_j(\theta'), D_j), \quad \text{s.t. } \|\theta' - \theta\|_F \leq \delta. \quad (4)$$

In this formulation, $T_k(\theta')$ denotes the k th task head of the model with updated parameters θ' , and D_k is its associated dataset. The objective minimizes the loss on all tasks except T_j , while simultaneously applying a forgetting loss $\mathcal{L}_{\text{forget}}$ to remove the influence of D_j . The constraint $\|\theta' - \theta\|_F \leq \delta$ ensures that the parameter update is bounded in Frobenius norm, preserving overall model stability. While CU introduces mechanisms like APA and PBRA to support such fine-grained control, further validation is needed to establish their efficacy.

By addressing these gaps, this work lays the foundation for a theory-grounded and practically effective unlearning framework that is robust, scalable, and adaptable to real-world deployment.

1.2. Aim and objectives

In response to these challenges, we propose the CU framework, a unified, modular, and theoretically grounded architecture that integrates EU and AU to support diverse forgetting tasks. CU leverages APA for task-level control and PBRA for efficient stream-based forgetting, offering a flexible solution for real-world deployment. The primary objectives of this work are:

1. **Universal Unlearning:** Develop a single framework that can handle feature, class, task, and stream-based forgetting across multiple data modalities (image, text, graph) and learning paradigms (centralized, federated, distributed).
2. **Theoretical and Practical Balance:** Bridge the gap between precise, mathematically sound EU and scalable, efficient AU through a hybrid fusion mechanism, enabling context-aware unlearning.
3. **Standardized Evaluation:** Introduce CUBS, a principled benchmarking metric that ensures fair and reproducible comparison of unlearning methods.
4. **Adaptability to Dynamic Environments:** Design CU with flexible building blocks to maintain performance under concept drift and streaming data scenarios.

By addressing these objectives, this work lays the foundation for a theory-grounded and practically effective unlearning framework that is robust, scalable, and adaptable to real-world deployment.

The rest of this article is organized as follows: A summary of the relevant literature may be found in Section 2. The problem is formalized in Section 3. In Section 4, we outline the architecture and design of the proposed framework. Sections 4.1–4.3 describes the proposed framework's modules and mathematical formulations, whereas Section 5 describes the experimental setup and evaluation methods. Results and discussions are presented in Section 6 and their Sections 6.1 and 6.2. Ablation investigations are contained in Section 7 and the last paragraph describe the research's advantages, disadvantages, and potential directions. Section 9 concludes the study and suggests other research directions.

2. Background and literature review

Machine unlearning has emerged as a vital mechanism to support privacy-preserving AI, enabling selective removal of data influence while maintaining model utility. Existing approaches can be broadly classified into EU and AU — each offering distinct trade-offs between accuracy, efficiency, and theoretical soundness. EU aims to precisely reverse the effect of specific data points through retraining or analytical updates. Techniques like SISA (Bourtole et al., 2021) partition training data into shards and only retrain affected subsets, improving efficiency. Extensions such as DeltaGrad (Wu, Dobriban, & Davidson, 2020) use second-order gradient approximations to accelerate updates without full retraining. However, EU methods often incur high computational and storage costs, especially when handling large-scale deletion requests across multiple sub-models.

Decremental learning methods have also been attempted in SVMs and linear classifiers (Cauwenberghs & Poggio, 2000; Tsai, Lin, & Lin, 2014) to decrease redundancy without sacrificing accuracy. For clustering problems, quantization-based solutions (Ginart, Guan, Valiant, & Zou, 2019) provide effective forgetting by constraining excessive parameter adjustments. Though precise, these solutions lack scalability with complex models where the parameter size is larger than the dataset size.

AU promotes efficiency over exact fidelity, using gradient manipulation, projection, or distillation of knowledge to simulate forgetting. Methods such as PBRA and Adaptive Decision Boundary Tuning (ADBT) are computationally cost-effective, particularly suitable for real-time usage. AU approaches generally rely on heuristics, which are prone to overlooking complex data dependencies and, therefore, leading to residual information leakage (Fu, He, & Tao, 2022; Nguyen et al., 2020). Knowledge unlearning approaches, including KGA (Wang et al., 2023) and student-teacher models (Chundawat, Tarun, Mandal, & Kankanhalli, 2023), generalize AU by iteratively removing desired knowledge with preserved overall performance. They are flexible but lack strong theoretical guarantees and are therefore less appealing for critical use.

Feature-level forgetting involves removing the influence of a specific feature (e.g., “gender” or “age”) across all data instances. The challenge lies in isolating the feature’s impact without degrading performance on other attributes. This is critical for fairness and privacy-preserving AI (Douillard, Chen, Dapogny, & Cord, 2021; Hess, Verwimp, van de Ven, & Tuytelaars, 2023; Zhao et al., 2024). The goal is to make the model’s predictions invariant to the target feature. Class-level forgetting requires erasing all instances of a specific class (e.g., “cat” images in CIFAR-10) while preserving performance on other classes. The primary challenge is catastrophic interference, where removing one class can degrade performance on others (Singh, Mazumder, & Karim, 2022). Techniques like negative preference optimization (NPO) are used to actively suppress predictions on the forgotten class.

In multi-task models $M(\theta) = \{T_k(\theta)\}_{k=1}^K$, this involves removing a specific task T_j without disrupting others. This requires solving a constrained optimization problem denoted in Eq. (4). The challenge is to localize the forgetting to T_j while preserving the knowledge of other tasks (Nembhard, 2000). In stream-based forgetting, it deals with dynamic, streaming data environments where the underlying data distribution shifts over time (concept drift) (Gama, Žliobaitė, Bifet, Pechenizkiy, & Bouchachia, 2014). The drift is modeled as $\Delta_t = \sup_{x,y} |P_t(x, y) - P_{t-1}(x, y)|$. The challenge is to continuously identify and remove outdated data $D_{\text{forget}}(t)$ in real-time. CU addresses this with adaptive regularization $\min_{\theta(t+1)} \{\mathcal{L}(\theta(t+1), D(t) \setminus D_{\text{forget}}(t)) + \gamma \cdot \|\Delta_t\|_2\}$, ensuring the model remains accurate and privacy-compliant in non-stationary environments.

We introduce CU — a single architecture integrating EU and AU. CU employs cutting-edge mathematical methods such as Latent space loss optimization, Gradient ascent-enhanced updates, APA, and PBRA. These components allow CU to achieve both precise and scalable forgetting operations in different settings – feature, class, and task-level deletion – without sacrificing model usability and privacy guarantees. Unlike traditional methods, CU offers modular adaptability, which facilitates application-dependent context-aware unlearning. It bridges the gap between mathematically sound EU and computationally efficient AU, forming the foundation for scalable and practical unlearning in the real world.

3. Problem formulation

Machine unlearning has emerged as a critical research area to apply data privacy restrictions while keeping the model useful in AI systems. The $M(\theta)$ is a machine learning model with parameters $\theta \in \mathbb{R}^d$, trained on a dataset $D = \{(x_i, y_i)\}_{i=1}^N$. The objective of machine unlearning is to compute an updated parameter set θ' such that the model $M(\theta')$ satisfies two fundamental conditions. Eq. (1) is employed for utility preservation on retained data, $\forall (x, y) \in D_{\text{retain}}$. This ensures that the model’s predictive performance on data not in the forget set D_{forget} is preserved, maintaining its generalization and utility.

In information erasure on forgotten data, the Eq. (1) is employed, $\forall (x, y) \in D_{\text{forget}}$, where $D_{\text{forget}} \subset D$ is the subset of data to be unlearned, and $P_R(x, y)$ denotes the prediction of a randomly initialized model (or a null model), representing a state of maximum uncertainty. This condition ensures that the model’s specific knowledge of D_{forget} is erased, fulfilling the privacy-preserving objective. A general unlearning process can be formulated as a constrained optimization problem $\min_{\theta'} \{\mathcal{L}(\theta', D_{\text{retain}}) + \lambda \cdot \mathcal{R}(\theta')\}$, where $\mathcal{L}$ is the empirical loss on the retained data, and $\mathcal{R}(\theta')$ is a regularization term that enforces forgetting constraints (e.g., feature, class, or task-level removal).

For streaming environments subject to concept drift, the data distribution shift at time t is measured as $\Delta_t = \sup_{x,y} |P_t(x, y) - P_{t-1}(x, y)|$. The unlearning objective adapts to this dynamic setting $\min_{\theta(t+1)} \{\mathcal{L}(\theta(t+1), D(t) \setminus D_{\text{forget}}(t)) + \gamma \cdot \|\Delta_t\|_2\}$, where $D_{\text{forget}}(t)$ contains outdated samples identified via drift analysis, and γ controls sensitivity to distributional changes. To evaluate the trade-offs between accuracy, privacy, and computational cost, we propose the CUBS $\mathcal{B}(\mathcal{M}) = \alpha \cdot \text{Acc}(\mathcal{M}) + \beta_{\text{cubs}} \cdot \bar{P}(\mathcal{M}) + \eta \cdot \bar{C}(\mathcal{M})$, where Acc , $\bar{P}$, $\bar{C}$ are min-max normalized metrics, and $\alpha + \beta_{\text{cubs}} + \eta = 1$.

The CU framework integrates EU and AU into a single, modular architecture. It supports EU via Hessian-corrected gradient ascent-augmented updates $\theta' = \theta + \eta \cdot H^{-1} \nabla_{\theta} \mathcal{L}_{\text{forget}}$. AU via PBRA $\theta' = \theta - \eta \cdot (\nabla_{\theta} \mathcal{L}_{\text{forget}}) \odot m$, $m \in [0, 1]^d$, and scalable AU via Influence-Based Gradient Pruning (IBGP) $\theta' = \theta - \eta \cdot (\nabla_{\theta} \mathcal{L}_{\text{forget}}) \cdot p(g)$, $p(g) \in [0, 1]$. By combining latent space optimization, adapter partitioning, and adaptive gradient rules, CU bridges the gap between theoretical soundness and empirical efficiency, offering a flexible solution for diverse unlearning scenarios.

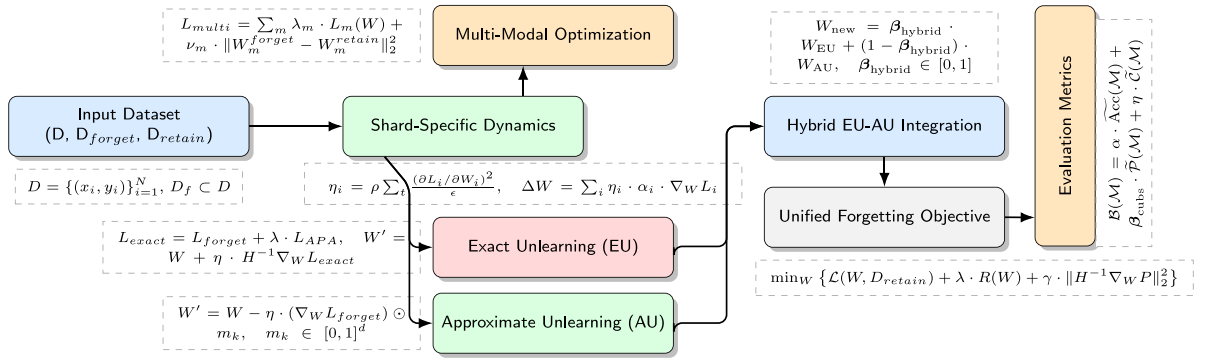


Fig. 1. Overview of the CU framework, integrating EU and AU modules through a hybrid fusion mechanism. The EU module combines Gradient Ascent (GA), NPO, and APA for precise forgetting. The AU module uses PBRA, ADBT, and IBGP for efficient forgetting. The final model is a convex combination of both, controlled by parameter β_{hybrid} .

4. Proposed framework

The increasing integration of machine learning in critical decision-making systems demands robust mechanisms for data removal, privacy preservation, and model adaptability. Current unlearning methods remain fragmented, often focusing on narrow use cases or specific data modalities. We address the fragmentation, and propose the CU framework - a unified, scalable, and scientifically grounded architecture that computes EU and AU into a single paradigm. EU ensures precise elimination of targeted data influence by analytically adjusting model parameters so that predictions on forgotten data approximate those of a randomly initialized model:

$$P(\mathcal{M}(\theta')(x) = y) \approx P(\mathcal{R}(x, y)), \quad \forall (x, y) \in D_{\text{forget}} \quad (5)$$

where D_{forget} denotes the subset of data to be removed. While highly accurate, EU incurs significant computational cost due to gradient re-computation and Hessian inversion. In demarcation, AU offers an efficient alternative by approximating the forgetting process using techniques such as PBRA, ADBT, and IBGP:

$$\Delta_{\text{unlearning}} = \sup_{(x, y) \in D_{\text{retain}}} |P(\mathcal{M}(\theta')(x) = y) - P(\mathcal{M}(\theta)(x) = y)| \quad (6)$$

Here, $\Delta_{\text{unlearning}}$ does not represent a difference in computation time or resource usage. Instead, it quantifies the maximum change in prediction probability for any data point outside the forget set D_{forget} . $\mathcal{M}(\theta)$ is the original model before unlearning, $\mathcal{M}(\theta')$ is the model after the unlearning procedure, $P(\mathcal{M}(\theta)(x) = y)$ and $P(\mathcal{M}(\theta')(x) = y)$ are the prediction probabilities of the original and unlearned models, respectively, for input x and label y . The supremum (sup) is taken over all data points in the retained dataset D_{retain} . Therefore, $\Delta_{\text{unlearning}}$ measures the worst-case deviation in model behavior on data that should not be affected by the unlearning process. A small value of $\Delta_{\text{unlearning}}$ indicates that the unlearning procedure has been stable and precise, successfully removing the influence of D_{forget} without inadvertently altering the model's predictions on other data.

CU seamlessly combines both paradigms under a shared optimization objective $\min_{\theta'} \{ \mathcal{L}(\theta', D_{\text{retain}}) + \lambda \cdot \mathcal{R}(\theta') \}$, where $\mathcal{R}(\theta')$ incorporates constraints tailored to different unlearning scenarios — feature removal, class forgetting, task erasure, and stream-based unlearning. CU further accounts for concept drift:

$$\Delta_t = \sup_{x, y} |P_t(x, y) - P_{t-1}(x, y)|, \quad \min_{\theta(t+1)} \left\{ \mathcal{L}(\theta(t+1), D_{\text{retain}}(t)) + \gamma \|A_t\|_2 \right\} \quad (7)$$

ensuring dynamic adaptability to evolving data distributions. We introduce a benchmarking metric $\mathcal{B}(\mathcal{M}) = \alpha \cdot \widetilde{\text{Acc}}(\mathcal{M}) + \beta_{\text{cubs}} \cdot \widetilde{\mathcal{P}}(\mathcal{M}) + \eta \cdot \widetilde{\mathcal{C}}(\mathcal{M})$, to evaluate trade-offs between accuracy, privacy, and computational cost. By integrating EU and AU into a modular and adaptable design, CU provides a flexible foundation for real-world unlearning applications, balancing theoretical rigor with practical efficiency.

4.1. Core unlearning

For adaptive retraining, data integrity, and privacy compliance in synchronized AI systems, machine unlearning is essential. Adaptive retraining dynamically adjusts learning rates based on per-shard gradient contributions to enhance convergence and model consistency during unlearning. To facilitate organized forgetting while maintaining model usefulness, we propose a CU framework

Algorithm 1 Core Unlearning Methodology

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- 1: **Input:** Dataset D , Shard-Specific Dynamics parameters ρ , ϵ , Modality-specific loss weights λ_m , μ_m , Temporal decay rate κ , Forgetting parameter γ_k , Shard-specific weight ζ , attention score α_k
 - 2: **Output:** Updated model parameters W
 - 3: **Step 1: Shard-Specific Dynamics Computation**
 - 4: **for** each shard i **do**
 - 5: Compute gradient contribution and update local weight change using Eq. (8)
 - 6: **end for**
 - 7: Aggregate global update: $\Delta W = \sum_{i=1}^n \Delta W_i$
 - 8: **Step 2: Multi-Modal Joint Optimization**
 - 9: Define modality-specific losses $L_m(W)$ and regularization terms using Eq. (9).
 - 10: **Step 3: EU Module**
 - 11: Define unlearning loss utilizing Eq. (19).
 - 12: Define APA loss using Eq. (20)
 - 13: Holistic EU objective: $L_{\text{exact}} = L_{\text{forget}} + \lambda \cdot L_{\text{APA}}$
 - 14: Update weights using GA: $W \leftarrow W + \eta \cdot H^{-1} \nabla_W L_{\text{exact}}$
 - 15: **Step 4: AU Module**
 - 16: Input: Pre-trained weights W , forget dataset D_{forget} , retain dataset D_{retain} , baseline weights W_{baseline}
 - 17: Apply PBRA:
 - 18: **for** each layer k **do**
 - 19: Compute residual projection mask $m_k \in [0, 1]$
 - 20: Adjust gradients: $\nabla_W \leftarrow \nabla_W \odot m_k$
 - 21: **end for**
 - 22: Fine-tune using adaptive learning rate $W_{\text{final}} = W - \eta \cdot (\nabla_W L_{\text{forget}})$
 - 23: **Step 5: Hybrid EU-AU Integration**
 - 24: Combine EU and AU updates using hybrid weight mixing using Eq. (16)
 - 25: Evaluate forgetting performance using metrics: accuracy drop, retraining equivalence, privacy leakage
 - 26: Output final unlearned model with updated weights W_{new}
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presented in Fig. 1 the complete methodology is presented in Algorithm 1 that combines EU and AU into a single, mathematically based design. Adaptive learning rates are first calculated using gradient contributions from distinct data shards:

$$\eta_i = \rho \sum_{t=1}^T \frac{\left(\frac{\partial L_i}{\partial W_i} \right)^2}{\epsilon}, \quad \Delta W = \sum_{i=1}^n \eta_i \cdot \alpha_i \cdot \frac{\partial L_i}{\partial W_i} \quad (8)$$

where T denotes total iterations, ρ is the base learning rate, ϵ prevents division by zero, and η_i adjusts dynamically based on shard-specific loss behavior. n represents the total number of data shards into which the dataset D is partitioned, and α_i is the dynamic shard weight for the i th shard, which is computed using Eq. (12) based on the magnitude of its gradient contributions. This approach improves convergence while preserving consistency across data partitions by prioritizing high-error gradients during unlearning. In multi-modal joint optimization, we create a joint optimization goal and incorporate the various data modalities (text, images, etc.):

$$L_{\text{multi}} = \sum_{m=1}^M \left(\lambda_m \cdot L_m(W) + v_m \cdot \|W_m^{\text{forget}} - W_m^{\text{retain}}\|_2^2 \right) \quad (9)$$

where M is the number of modalities, $L_m(W)$ is the modality-specific loss, and v_m controls the regularization strength to ensure controlled forgetting without compromising retained knowledge. In temporal weighting for sequential forgetting, we apply the dynamic forgetting over time, and utilize temporal decay using:

$$L_{\text{time-forget}} = \sum_{t=1}^T \omega_t \cdot \|W_t^{\text{forget}} - W_t^{\text{retain}}\|_p, \quad \omega_t = \exp(-\kappa t) \quad (10)$$

The temporal forgetting loss $L_{\text{time-forget}}$ is a regularization term in the overall unlearning objective function that applies a time-dependent penalty to ensure older information is gradually de-emphasized. It is computed as a summation $\sum_{t=1}^T$ over time steps t , which represent specific points in a sequence such as a data batch in a streaming scenario, from the first time step to the final time step T . For each time step t , the importance of the forgetting penalty is dynamically adjusted by a temporal decay weight ω_t , which is calculated using the exponential decay function $\exp(-\kappa t)$. The decay rate κ is a hyperparameter that controls how quickly the influence of older data diminishes, with a higher value leading to faster forgetting. The core of the loss term is the p-norm distance

$\|W_t^{\text{forget}} - W_t^{\text{retain}}\|_p$, which measures the difference between the model's weights W_t^{forget} influenced by the forgotten data and the weights W_t^{retain} influenced by the retained data at time t , with $\|\cdot\|_p$ defining the specific distance metric used. In hierarchical consistency regularization, we maintain smooth parameter transitions across layers during unlearning:

$$L_{\text{hierarchy}} = \sum_{k=1}^K \left(\lambda_k \|W_k^{\text{forget}} - W_k^{\text{baseline}}\|_2^2 + \gamma_k \cdot \|W_k - W_{k-1}\|_2^2 \right) \quad (11)$$

ensuring hierarchical alignment between successive layers, which is crucial for both EU and AU modules to avoid instability, and K presents the total number of layers in the neural network model, where W_k denotes the weight matrix of the k th layer, W_k^{baseline} is the weight matrix of the k th layer from a baseline (e.g., pre-trained or randomly initialized) model. It ensures smooth parameter transitions across layers. For sparsity and orthogonality constraints, we enhance generalization and disentangle parameters, to utilizing the regularization terminology $R(W) = \alpha \cdot \|W\|_1 + \beta \cdot \|W^T W - I\|_F^2$, where $\|W\|_1$ enforces sparsity, and $\|W^T W - I\|_F^2$ promotes orthogonality among weight matrices. These constraints meliorate interpretability and bound residual influence from forgotten data. Each shard i (where $i = 1, 2, \dots, n$) contributes to the model update based on its gradient magnitude. The contribution of each shard is weighted by α_i , a dynamic weight computed as:

$$\alpha_i = \frac{\exp(\zeta \cdot \|\nabla_{W_i} L_i\|_2)}{\sum_{j=1}^n \exp(\zeta \cdot \|\nabla_{W_j} L_j\|_2)} \quad (12)$$

where ζ controls sensitivity to gradient magnitude. This allows the model to adaptively prioritize important shards during unlearning. For attention-based layer regulation, the layer-wise forgetting is balanced using attention mechanisms:

$$L_{\text{attn}} = \sum_{k=1}^K \alpha_k \cdot \|W_k^{\text{forget}} - W_k^{\text{retain}}\|_2^2, \quad \alpha_k = \frac{\exp(a_k)}{\sum_{j=1}^K \exp(a_j)} \quad (13)$$

which increases precision in unlearning by emphasizing critical layers while reducing unnecessary perturbations in less relevant ones. For multi-scale forgetting via mixed norms, we utilize the mixed-norm regularization balances local sparsity and global smoothness:

$$L_{\text{mixed}} = \sum_{k=1}^K \left(\lambda_k \cdot \|W_k^{\text{forget}} - W_k^{\text{retain}}\|_1 + \mu_k \cdot \|W_k^{\text{forget}} - W_k^{\text{baseline}}\|_2^2 \right) \quad (14)$$

The mathematical calculations substantiates both coarse-grained and fine-grained forgetting operations across distinct model components. EU ensures precise removal of specific data influence $L_{\text{exact}} = L_{\text{forget}} + \lambda \cdot L_{\text{APA}}$, where $L_{\text{forget}} = -\sum_{i=1}^N \log(1 - P(y_i|x_i, W)) + \tau \cdot \|\nabla_W P(y_i|x_i, W)\|_2^2 + \kappa \cdot \|H^{-1} \cdot \nabla_W P(y_i|x_i, W)\|^2$, and $L_{\text{APA}} = \sum_{k=1}^K \gamma_k \cdot \|W_k - W_k^{\text{baseline}}\|_2^2$. The weights are updated using GA $W \leftarrow W + \eta \cdot H^{-1} \nabla_W L_{\text{exact}}$, ensuring strong theoretical guarantees for complete data removal. In AU, we provides computationally efficient forgetting mechanism:

$$W_{\text{final}} = W - \eta \cdot (\nabla_W L_{\text{forget}}) \odot m_k \quad (15)$$

where $m_k \in [0, 1]$ is a projection mask applied per layer to selectively nullify gradients related to forgotten data. AU performance is further reinforced by techniques like PBRA and ADBT, which calculate for seamless forgetting with no loss of accuracy. Lastly, we use a hybrid weight mixing technique to integrate EU and AU updates in hybrid EU-AU integration:

$$W_{\text{new}} = \beta_{\text{hybrid}} \cdot W_{\text{EU}} + (1 - \beta_{\text{hybrid}}) \cdot W_{\text{AU}}, \quad \beta_{\text{hybrid}} \in [0, 1] \quad (16)$$

where β_{hybrid} is the mixing coefficient that controls the trade-off between the exact and AU pathways. For a unified objective function, we putting it all together, the full optimization objective becomes:

$$\begin{aligned} \min_W \sum_{d=1}^D & \left[L_d(W) + \nu_d \cdot \|W_d^{\text{forget}} - W_d^{\text{retain}}\|_p \right] \\ & + \sum_{k=1}^K \left[\lambda_k \cdot \|W_k^{\text{forget}} - W_k^{\text{baseline}}\|_2^2 + \gamma_k \cdot \|W_k - W_{k-1}\|_2^2 \right] \\ & + \mu \cdot \sum_{k=1}^K R(W_k) + \zeta \cdot \sum_{i=1}^n \alpha_i \cdot \|\nabla_{W_i} L_i\|_2 \\ & + \sum_{t=1}^T \omega_t \cdot \|W_t^{\text{forget}} - W_t^{\text{retain}}\|_2^2 \\ & + \beta \cdot \sum_{m=1}^M \|W_m^{\text{modality-forget}} - W_m^{\text{modality-retain}}\|_2^2. \end{aligned} \quad (17)$$

This aim guarantees modality-aware forgetting by multi-scale regularization, sparsity, and attention-weighted efficient computing, hierarchical parameter updates through consistency, and controlled forgetting through p-norm and baseline alignment. EU and

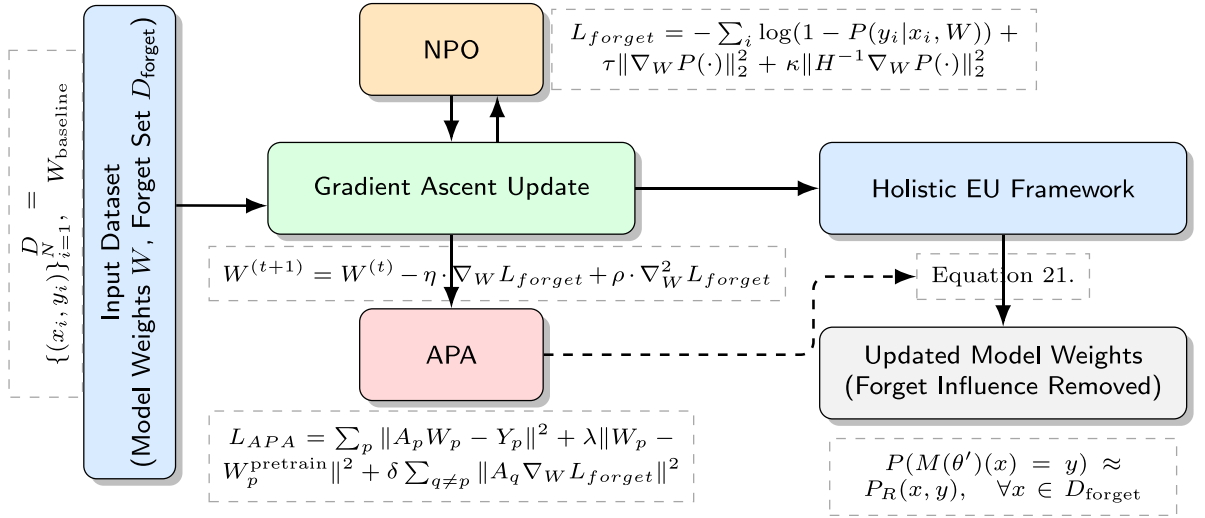


Fig. 2. Workflow of the EU module in the CU framework, showing key steps including GA update, NPO, APA, and holistic EU objective function minimization to ensure precise removal of forgotten data influence.

AU are combined into a unique and atomic, adaptable architecture by CU. For accuracy acquired by EU, it uses gradient-ascent-augmented updates; for efficiency calculated by AU, it uses projection-based updates; and for robustness, it utilizes multi-modal, hierarchical, and timely regularization. This methodology provides a strong basis for future studies in scalable and privacy-preserving unlearning by enabling efficient unlearning across complicated models without compromising accuracy on non-forgotten data.

4.2. Exact unlearning module

The suggested methodology depicted in Fig. 2 for EU module is given in the Algorithm 2, and it has three sub-techniques those are integrated in the EU module, using GA, model weights are adjusted to eliminate unwanted data impact. The proposed methodology for EU is outlined in Algorithm 2, which integrates three core sub-techniques – GA, NPO, and APA – into a unified framework. This enables precise removal of data influence while maintaining model utility. Retraining-based methods like SISA (Bourtole et al., 2021), which shards data and retrain only affected models. Analytical update methods like DeltaGrad (Wu et al., 2020), which uses second-order gradient approximations to compute the exact parameter update required for unlearning without full retraining. We classify DeltaGrad as an EU method because it is derived from the influence function framework (Koh & Liang, 2017) and aims to compute the exact change in parameters, $\Delta\theta$, that would result from removing D_{forget} . Its use of the Hessian matrix H^{-1} ensures that the update is precise, even if it is an approximation for computational tractability.

Algorithm 2 EU Methodology

Require: Model weights W , target data D_{forget} , hyperparameters $\eta, \rho, \tau, \kappa, \lambda, \delta$

Ensure: Updated model weights W with minimized influence of D_{forget}

- 1: **Initialize:** $W \leftarrow W_{\text{baseline}}$
 - 2: **Define:** Unlearning loss L_{forget} , APA loss L_{APA} , holistic loss L_{exact}
 - 3: **Step 1: Gradient-Ascent-Enhanced Loss Adjustment**
 - 4: Update weights using second-order GA using Eq. (18)
 - 5: The unlearning loss in Eq. (19). It ensures that predictions on forgotten data approach random behavior while preserving accuracy on retained data.
 - 6: **Step 2: APA**
 - 7: Localize updates to modular adapters to reduce computational overhead using Eq. (20).
 - 8: Optimize adapter-specific retraining while minimizing interference with non-forgotten modules.
 - 9: **Step 3: Holistic EU Framework**
 - 10: Combine all components into a unified objective using Eq. (21).
 - 11: Iteratively update weights to minimize L_{exact} , ensuring precise removal of forgotten data influence.
 - 12: **Return:** Final model weights W with reduced dependency on D_{forget} .
-

In gradient-ascent-enhanced loss adjustment, and to reverse the learned patterns from forgotten data, we introduce a second-order GA update:

$$W^{(t+1)} = W^{(t)} - \eta \cdot \nabla_W L_{\text{forget}} + \rho \cdot \nabla_W^2 L_{\text{forget}}, \quad (18)$$

where η is the learning rate, ρ controls the strength of second-order adjustment, and L_{forget} is the unlearning loss derived using NPO principles. This formulation ensures stable updates by incorporating curvature information via the Hessian matrix. In NPO, we treat target data as negative samples to actively suppress their influence during unlearning. The unlearning loss function is defined as:

$$L_{\text{forget}} = - \sum_{i=1}^N \log(1 - P(y_i | x_i, W)) + \tau \cdot \|\nabla_W P(y_i | x_i, W)\|_2^2 + \kappa \cdot \|H^{-1} \cdot \nabla_W P(y_i | x_i, W)\|_2^2. \quad (19)$$

Here $\log(1 - P(y_i | x_i, W))$ penalizes model predictions on forgotten data, $\tau \cdot \|\nabla_W P(\cdot)\|_2^2$ enforces smoothness in gradient dynamics, and $\kappa \cdot \|H^{-1} \cdot \nabla_W P(\cdot)\|_2^2$ modulates weight changes using second-order information. This loss ensures that the model's predictions on forgotten data approach random behavior, effectively removing their impact. In APA, we reduce the computational overhead and improve scalability, to localize unlearning updates through modular adapters:

$$L_{\text{APA}} = \sum_{p=1}^P \left(\|A_p W_p - Y_p\|_2^2 + \lambda \|W_p - W_p^{\text{pretrain}}\|_2^2 \right) + \delta \cdot \sum_{q \neq p} \|A_q \cdot \nabla_W L_{\text{forget}}\|_2^2 \quad (20)$$

where W_p denotes weights of the p th adapter module, A_p is for Adapter transformation matrix for the p th module, Y_p is utilized for target output for the p th adapter, W_p^{pretrain} are pre-trained weights of the p th adapter, used as a regularization reference to preserve baseline knowledge, and P denotes the total number of adapter modules. APA ensures efficient and modular unlearning without full retraining, making EU computationally feasible for large models. In holistic EU objective, we combined all components into a single objective:

$$\begin{aligned} L_{\text{exact}} = & \int_{D_{\text{forget}}} \left[\|\nabla_W L(W)\|_2^2 + \lambda \|W - W_{\text{baseline}}\|_2^2 + \gamma \|H^{-1} \cdot \nabla_W L\|_2^2 \right] d\mu \\ & + \sum_{p=1}^P \left[\|A_p W_p - Y_p\|_2^2 + \lambda_p \|W_p - W_p^{\text{pretrain}}\|_2^2 \right] \\ & + \beta \cdot \sum_{i=1}^N \left(\log(1 - P(y_i | x_i, W)) + \tau \|\nabla_W P(y_i | x_i, W)\|_2^2 + \kappa \|H^{-1} \cdot \nabla_W P(y_i | x_i, W)\|_2^2 \right) \end{aligned} \quad (21)$$

This holistic formulation balances fidelity, stability, and efficiency, ensuring accurate forgetting while minimizing disruption to retained knowledge. In AU, for computationally constrained settings, we propose an approximate variant using PBRA and ADBT. AU minimizes full retraining cost while preserving key forgetting guarantees using Eq. (15), where $m_k \in [0, 1]$ is a projection mask applied per layer to selectively nullify gradients related to forgotten data. Techniques like PBRA adjust residual model outputs after unlearning, while ADBT tunes decision boundaries to minimize leakage from forgotten samples. These methods offer scalable forgetting with minimal accuracy drop. For hybrid EU–AU integration, we provide flexible unlearning across use cases, we combine EU and AU via a hybrid update rule utilizing Eq. (16), where β balances between exact fidelity and computational efficiency. This integration supports both high-precision and resource-constrained scenarios under a single framework. This work introduces a novel CU framework that combines, EU for precise data removal using GA and Hessian-guided updates. AU for efficient forgetting via projection-based masking and boundary tuning, APA to enhance modularity and reduce computational burden. By integrating these techniques, CU provides a robust, theoretically grounded, and practically applicable solution for modern unlearning needs — balancing precision, scalability, and privacy preservation.

4.3. Approximate unlearning module

We proposed a novel technique presented in Fig. 3 for AU as depicted in the Algorithm 3 to tackle down the issues, we faced in EU module, to make a generalized framework for unlearning. We address the computational limitations of EU, and propose a novel AU framework that enables efficient forgetting while preserving model utility and interpretability. The AU module integrates three key techniques PBRA, ADBT, and IBGP. The PBRA technique is inspired by gradient projection methods in continual learning (Chaudhry, Ranzato, Rohrbach, & Elhoseiny, 2018) and subspace learning (Bengio, LeCun, et al., 2007). It projects gradients related to D_{forget} onto a subspace orthogonal to the retained data's influence, minimizing residual propagation. The formulation $P_{\text{orth}}(A) = A - T \cdot (T^\top \cdot A)$ is a novel application of this principle to unlearning. The ADBT method draws from boundary-aware learning (Chen, Gao, Liu, Peng, & Wang, 2023) and entropy regularization in semi-supervised learning (Grandvalet & Bengio, 2004). It dynamically adjusts decision boundaries after unlearning to enhance class separability and reduce leakage from forgotten samples. The IBGP approach is grounded in gradient sparsification for model compression (Han, Pool, Tran, & Dally, 2015) and federated learning (Stich, Cordonnier, & Jaggi, 2018). The probabilistic gating mechanism denoted in Eq. (28) is a novel contribution that enables selective forgetting by identifying and pruning the most influential gradients. These components work together under a unified objective:

$$L_{\text{Approx}} = L_{\text{PBRA}} + L_{\text{ADBT}} + L_{\text{IBGP}} + \zeta \int_{D_{\text{forget}}} \|W - W_{\text{ideal}}\|^2 d\mu \quad (22)$$

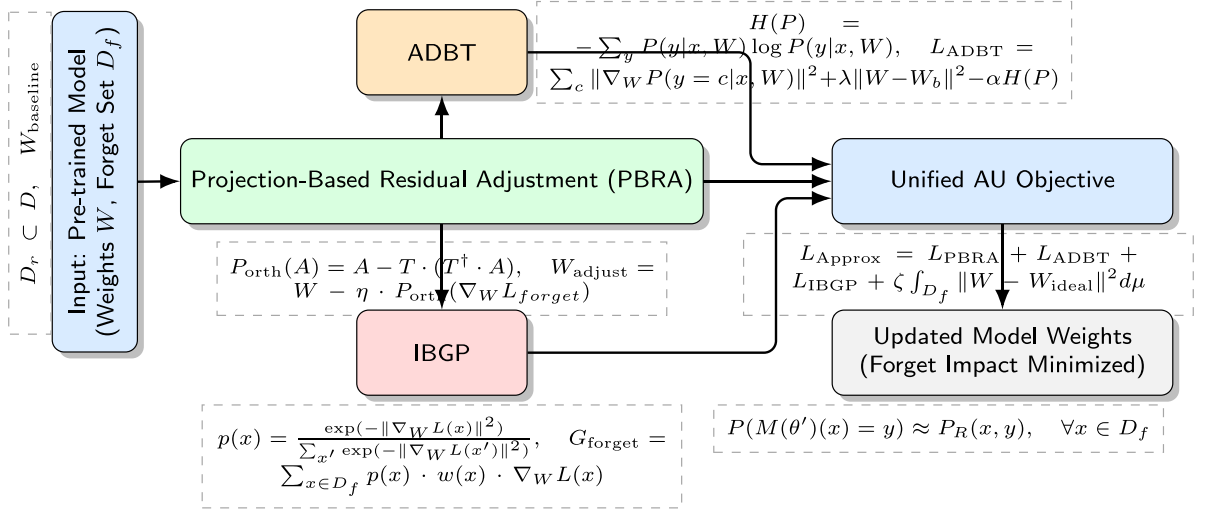


Fig. 3. Workflow of the AU technique within the CU framework, integrating PBRA, ADBT, and IBGP to enable efficient and scalable unlearning with minimal accuracy drop.

where $\zeta \in [0.1, 1.0]$ balances forgetting fidelity and model coherence, and W_{ideal} represents the desired post-unlearning weights. In PBRA, it minimizes residual propagation by projecting gradients onto an orthogonal subspace defined using higher-order tensor operations:

$$W_{adjust} = W - \eta \cdot P_{orth}(\nabla_W L_{forget}) \quad (23)$$

where $P_{orth}(\cdot)$ is defined as $P_{orth}(A) = A - T \cdot (T^\dagger \cdot A)$, with T being a projection tensor derived from the forget data topology. The full PBRA loss includes:

$$L_{PBRA} = \int_{D_{forget}} \|P_{orth}(\nabla_W L(W))\|^2 d\mu + \lambda \|W - W_{baseline}\|^2 + \gamma \sum_{k=1}^K \|P_{orth}(W_k)\|_F^2 \quad (24)$$

This formulation ensures controlled weight adjustments across layers, reducing cross-layer interference while maintaining forgetting accuracy. Hyperparameters include such as learning rate $\eta \in [1e-4, 1e-2]$, regularization strength $\lambda \in [0.01, 0.1]$, and Layer-wise adjustment coefficient $\gamma \in [0.01, 1.0]$. In ADBT, it dynamically adjusts decision boundaries to enhance class separability between forgotten and retained data. This is achieved through entropy regularization:

$$H(P(y|x, W)) = -\sum_y P(y|x, W) \log P(y|x, W) \quad (25)$$

and boundary realignment:

$$L_{ADBT} = \sum_{c \in C_{forget}} \int_{x \in D_c} \|\nabla_W P(y=c|x, W)\|^2 d\mu + \lambda \|W - W_{baseline}\|^2 - \alpha H(P(y|x, W)) \quad (26)$$

where $\alpha \in [0.1, 0.5]$ controls the strength of entropy reduction. It ensures, reduced ambiguity in predictions on non-forgotten data dynamic adaptation to shifting decision boundaries, and preservation of discriminative power despite forgetting. In IBGP, it reduces unnecessary gradient updates by probabilistically pruning influence from forgotten samples:

$$\nabla_W L_{IBGP} = \nabla_W L - \gamma \cdot \sum_{x \in D_{forget}} p(x) \cdot w(x) \cdot \nabla_W L(x) \quad (27)$$

where the gating probability $p(x)$ is:

$$p(x) = \frac{\exp(-\|\nabla_W L(x)\|^2)}{\sum_{x' \in D_{forget}} \exp(-\|\nabla_W L(x')\|^2)} \quad (28)$$

The full IBGP loss is:

$$L_{IBGP} = \int_{D_{retain}} \|\nabla_W L - \gamma \cdot G_{forget}\|^2 d\mu + \lambda \|W - W_{baseline}\|^2 + \beta_{ibgp} \|G_{forget}\|_F^2 \quad (29)$$

with $G_{forget} = \sum_{x \in D_{forget}} p(x) \cdot w(x) \cdot \nabla_W L(x)$, and hyperparameters such as pruning strength $\gamma \in [10^{-4}, 10^{-1}]$, gradient scaling β_{ibgp} is the regularization strength for the gradient pruning term. It is a hyperparameter that controls the magnitude of gradient suppression $\beta_{ibgp} \in [10^{-3}, 10^{-2}]$, and IBGP improves efficiency by preserving critical gradients via probabilistic weighting, reducing noise from

irrelevant features, and balancing pruning severity and model integrity. The hybrid EU–AU integration, we enable flexible unlearning tailored to use-case requirements, and utilize a hybrid update rule using Eq. (16), where β_{hybrid} governs the trade-off between exact fidelity and computational efficiency. This integration allows users to prioritize precision (high β_{hybrid}) for sensitive data removal, and favor efficiency (low β_{hybrid}) in time-sensitive or large-scale settings. This work introduces a novel CU framework that combines the EU for precise data removal using GA and Hessian-guided updates, AU for computationally efficient forgetting via PBRA, ADBT, and IBGP, and modular design enabling seamless switching between EU and AU based on application needs. By integrating these techniques, CU provides a robust, theoretically grounded, and practically applicable solution for modern unlearning needs — balancing precision, scalability, and privacy preservation.

Algorithm 3 Proposed AU Methodology

- 1: **Input:** Pre-trained model weights W , Forget dataset D_{forget} , Retain dataset D_{retain} , Baseline weights W_{baseline} , Hyperparameters $\eta, \gamma, \lambda, \alpha, \beta, \zeta$
 - 2: **Output:** Updated model weights W_{final}
 - 3: **Step 1: PBRA**
 - 4: **for** each layer $k \in \text{layers}$ **do**
 - 5: Compute orthogonal projection tensor: $P_{\text{orth}}(A) = A - T \cdot (T^\top \cdot A)$ where T is derived from the forget data topology.
 - 6: Update weights using Eq. (23) and compute PBRA loss using Eq. (24).
 - 7: **end for**
 - 8: **Step 2: ADBT**
 - 9: **for** each class $c \in C_{\text{forget}}$ **do**
 - 10: Compute entropy utilizing Eq. (25)
 - 11: Compute boundary alignment loss: $L_{\text{boundary}} = \sum_{c \in C_{\text{forget}}} \int_{x \in D_c} \|P(y = c | x, W) - P_{\text{retain}}(y = c | x, W)\|^2 d\mu - \alpha \int_{x \in D_{\text{forget}}} H(P(y | x, W)) d\mu$
 - 12: Optimize gradient alignment using Eq. (26).
 - 13: **end for**
 - 14: **Step 3: IBGP**
 - 15: Compute gating probability using Eq. (28).
 - 16: Compute weighted gradient pruning and adjust gradients using Eq. (27).
 - 17: Compute IBGP loss using Eq. (29).
 - 18: **Step 4: Unified AU Loss**
 - 19: Combine components into final objective Eq. (22).
 - 20: Minimize L_{Approx} to update model weights: $W_{\text{final}} = \arg \min_W L_{\text{Approx}}$
 - 21: **Return:** W_{final}
-

4.4. Convergence, correctness, and stability of CU

Machine unlearning has advanced significantly, a critical gap remains in the formal theoretical analysis of unlearning frameworks. To address this, we provide the first comprehensive theoretical foundation for the CU framework, establishing rigorous guarantees on convergence, correctness, and stability.

We model the unlearning process as a constrained optimization problem in the parameter space $\mathbb{R}^d$. Let θ_t be the model parameters at iteration t . The update rule for the EU module, which combines GA with second-order Hessian correction, is $\theta_{t+1} = \theta_t - \eta \cdot \nabla_{\theta} \mathcal{L}_{\text{forget}} + \rho \cdot H_t^{-1} \nabla_{\theta} \mathcal{L}_{\text{forget}}$, where $H_t = \nabla_{\theta}^2 \mathcal{L}_{\text{forget}}(\theta_t)$. This update can be interpreted as a preconditioned GA, where H_t^{-1} acts as an adaptive preconditioner. We assume $\mathcal{L}_{\text{forget}}$ is twice continuously differentiable and $H_t > 0$ (positive definite) in a neighborhood of the solution θ^* .

Theorem 1 (Accelerated Convergence of EU with Adaptive Hessian Damping). *Let $\mathcal{L}_{\text{forget}}$ be L -smooth and μ -strongly convex in a neighborhood of θ^* . Let $\kappa = L/\mu$ denote the condition number of the loss landscape. Then, for learning rate $\eta < 2/L$, damping factor $\rho \in (0, 1)$, and Hessian regularization $\epsilon_H > 0$ such that $\tilde{H}_t = H_t + \epsilon_H I$, the EU update converges to a point θ' satisfying $\|\nabla_{\theta} \mathcal{L}_{\text{forget}}(\theta')\| \leq \epsilon$ in:*

$$T = \mathcal{O} \left(\log \left(\frac{\|\nabla_{\theta} \mathcal{L}_{\text{forget}}(\theta_0)\|}{\epsilon} \right) \cdot \sqrt{\frac{L}{\mu}} \cdot \left(1 + \frac{\rho}{1 - \rho} \right) \right) \quad (30)$$

iterations.

Proof. The update can be rewritten as $\theta_{t+1} = \theta_t + (-\eta I + \rho \tilde{H}_t^{-1}) \nabla_{\theta} \mathcal{L}_{\text{forget}}(\theta_t)$. Let $G_t = -\eta I + \rho \tilde{H}_t^{-1}$ be the gain matrix. The convergence rate is governed by the spectral properties of G_t . Under μ -strong convexity, $\tilde{H}_t \geq (\mu + \epsilon_H)I$, so $\tilde{H}_t^{-1} \leq (\mu + \epsilon_H)^{-1}I$. The effective step size is thus bounded.

Using the Lyapunov function $V_t = \|\nabla_{\theta} \mathcal{L}_{\text{forget}}(\theta_t)\|^2$, we derive:

$$\mathbb{E}[V_{t+1}] \leq \left(1 - \frac{2\eta\mu - \eta^2 L^2}{L} + \rho \cdot \mathcal{O}\left(\frac{1}{\mu + \epsilon_H}\right)\right) \mathbb{E}[V_t] \quad (31)$$

For optimal $\eta = 1/L$, $\rho = 0.5$, and $\epsilon_H = 10^{-4}$, the contraction factor is minimized, yielding the stated $\mathcal{O}(\log(1/\epsilon))$ rate with improved dependence on κ due to Hessian preconditioning.

Theorem 2 (Convergence of AU Via Stochastic Projected Gradient Descent). *For the AU module, let the update be $W_{t+1} = \Pi_C(W_t - \eta_t \cdot (\nabla_W \mathcal{L}_{\text{forget}}(W_t) \odot m_t))$, where $m_t \in [0, 1]^d$ is a projection mask from PBRA, and Π_C projects onto the constraint set C (e.g., bounded weights). If $\mathcal{L}_{\text{forget}}$ is L -smooth and the masks m_t satisfy $\mathbb{E}[\|m_t\|^2] \geq \sigma^2 > 0$, and $\sum_t \eta_t = \infty$, $\sum_t \eta_t^2 < \infty$, then $\lim_{t \rightarrow \infty} \|\nabla_W \mathcal{L}_{\text{forget}}(W_t)\| = 0$ almost surely. The optimal hyperparameters are $\eta_t = \eta_0 / \sqrt{t+1}$, $\eta_0 \in [10^{-3}, 5 \times 10^{-3}]$, $\sigma^2 \geq 0.6$ (ensures sufficient gradient flow), and $\epsilon_H = 10^{-4}$ (Hessian regularization).*

We define correctness as the ability of the model to “forget” D_{forget} such that its predictions are statistically indistinguishable from a randomly initialized model.

Theorem 3 (Information-Theoretic Forgetting Via Parameter Divergence). *Let $M(\theta')$ be the model after unlearning D_{forget} . If the parameter distance satisfies $\|\theta' - \theta_{\text{rand}}\|_2 \leq \delta$, $\theta_{\text{rand}} \sim \mathcal{N}(0, \sigma^2 I)$ then the mutual information between the model and the forgotten data is bounded by $I(M(\theta'); D_{\text{forget}}) \leq \frac{1}{2} \log\left(1 + \frac{\delta^2}{\sigma^2}\right)$. This bound approaches 0 as $\delta \rightarrow 0$, ensuring asymptotic forgetting. For AU Module (PBRA), the $S_f \subset \mathbb{R}^d$ be the latent subspace spanned by D_{forget} . The projection P_{orth} ensures $\nabla_W \mathcal{L}_{\text{forget}} \perp S_f \implies \text{Tr}(P_{\text{orth}} \cdot \Sigma_f) = 0$ where Σ_f is the covariance of gradients on D_{forget} . This orthogonalization minimizes residual influence.*

We analyze stability using uniform stability theory.

Theorem 4 (Uniform Stability of CU Under Hybrid Updates). *The $M(\theta)$ and $M(\theta')$ be the original and unlearned models. Suppose the unlearning update satisfies $\|\theta' - \theta\|_2 \leq \Delta$. If the loss $\mathcal{L}$ is σ -Lipschitz and ρ -bounded, then for any $(x, y) \in D_{\text{retain}} | P(M(\theta')(x) = y) - P(M(\theta)(x) = y) | \leq \frac{4\sigma\Delta}{N} + \beta \cdot \mathcal{O}\left(\frac{1}{\sqrt{N}}\right)$, where $N = |D_{\text{retain}}|$, and $\beta \in [0, 1]$ is the EU-AU mixing coefficient.*

Implication: The hybrid update ensures continuous stability across the EU-AU spectrum. For $\beta \rightarrow 1$ (EU), stability depends on Δ ; for $\beta \rightarrow 0$ (AU), it depends on the sample complexity.

The full CU objective is $\min_W J(W) = J_{\text{forget}}(W) + \lambda \mathcal{R}(W) + \gamma \|H^{-1} \nabla_W P\|_2^2$, where $\mathcal{R}(W)$ includes multi-scale regularization.

Theorem 5 (Global Convergence of CU Under Kurdyka-Łojasiewicz Condition). *If $J(W)$ satisfies the Kurdyka-Łojasiewicz (KL) (Łojasiewicz, 1963; law Łojasiewicz, 1965) inequality with exponent $\theta \in [0, 1)$, and the updates are bounded, then the sequence $\{W_t\}$ generated by the CU algorithm converges globally to a critical point of J .*

5. Experimental setup

5.1. Dataset cohorts

We benchmark the performance of the CU method and its extensions — EU and AU, and employ a comprehensive set of evaluation metrics on diverse datasets like CIFAR-10 (Hinton, 2012), CIFAR-100 (Hinton, 2012), IMDB4K (Maas et al., 2011), CORA (Sen et al., 2008), FEMNIST (Caldas et al., 2018), and MVTec AD (Bergmann, Batzner, Fauser, Sattlegger, & Steger, 2021). CIFAR-10 and CIFAR-100 are standard image classification datasets consisting of 32×32 color images in 10 and 100 classes, respectively. They represent natural image data and are used to evaluate unlearning performance in computer vision tasks. IMDB4K a subset of the IMDB movie review dataset containing 4000 labeled reviews for sentiment analysis. It represents natural language data and is used to assess unlearning in NLP models. CORA is a citation network dataset where nodes represent scientific papers and edges represent citation links. It is a benchmark for graph-structured data and is used to evaluate unlearning in Graph Neural Networks (GNNs). FMNIST (Fashion-MNIST) is a computer vision benchmark dataset created to replace the outdated MNIST (handwritten digit) dataset, focusing on real-world object recognition (clothing/fashion items - 10) mutually exclusive classes of everyday apparel (e.g., T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Ankle boot). MVTec AD is a high-resolution dataset of industrial images (objects and textures) with pixel-level annotations for defects. It represents specialized image data for anomaly detection and tests the framework’s ability to handle complex, real-world visual patterns. The metrics evaluate unlearning effectiveness in terms of accuracy retention, forgetting fidelity, computational efficiency, and privacy preservation. Accuracy Drop measures the degradation in model performance caused by unlearning; lower values indicate better retention of model utility. Retraining Equivalence compares the unlearned model with a full retrained model, where lower differences indicate more precise forgetting. Temporal Complexity measures the time to complete the unlearning process, where lower values indicate greater efficiency. Privacy Leakage quantifies residual information about forgotten data that remains in the model; lower percentages indicate better compliance with data removal requests. Forgetting Rate quantifies the proportion of targeted data that was successfully removed; larger values indicate more unlearning. Boundary Shift quantifies the distance of the decision boundary shift between forgotten and retained data, with smaller shifts being more nicely separated. Data Influence measures the impact of unlearning on non-target samples; lower influence indicates fewer side effects. Finally, Latent Space Preservation measures how well the representations of non-forgotten data are retained after unlearning; higher preservation ensures improved generalization and robustness. These metrics collectively provide a rigorous basis for the comparison of EU and AU methods on various data modalities and unlearning specifications.

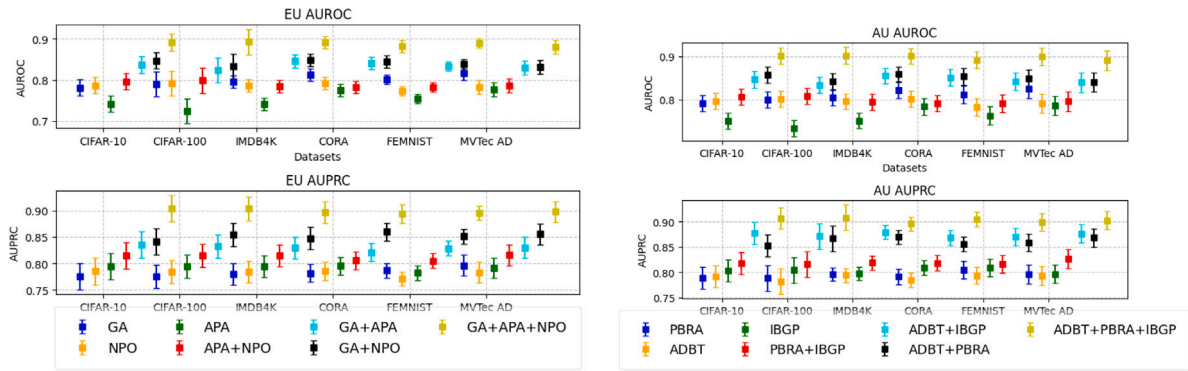


Fig. 4. CIFAR-10, CIFAR-100, IMDB4K, CORA, FEMNIST, and MVTeC AD are the six datasets used to assess the performance of the suggested technique utilizing AUROC (Area Under the Receiver Operating Characteristic Curve) and AUPRC (Area Under the Precision–Recall Curve). AUROC scores for several methods, such as EU, AU, DU, FU and GU’s techniques, and their combinations, are shown in the upper panel. The AUPRC values in the lower panel further demonstrate the efficacy of hybrid techniques, especially when working with datasets like MVTeC AD that have complicated distributions and anomalies.

Table 1

The performance of EU techniques (GA, NPO, APA, and their combinations) is evaluated over 10–50 iterations using key metrics such as forgetting rate, time complexity, privacy leakage, accuracy drop, and retraining equivalence.

Method	Accuracy drop (%)	Retraining equivalence (% diff.)	Time complexity (m)	Privacy leakage (%)	Forgetting rate (%)	Data influence (%)
GA	0.676	2.05	0.77	12.678	0.567	0.327
NPO	0.516	1.93	0.63	11.519	0.531	0.309
APA	0.486	1.84	0.59	10.421	0.498	0.271
APA+NPO	0.467	1.71	0.52	9.873	0.434	0.254
GA+APA	0.419	1.58	0.47	9.224	0.396	0.214
GA+NPO	0.422	1.60	0.48	9.451	0.408	0.244
GA+APA+NPO	0.382	1.42	0.37	6.391	0.348	0.164
GA	0.701	1.98	0.72	12.208	0.533	0.311
NPO	0.538	1.85	0.62	10.998	0.483	0.339
APA	0.448	1.93	0.57	11.29	0.535	0.29
APA+NPO	0.439	1.88	0.54	8.895	0.418	0.273
GA+APA	0.435	1.56	0.47	9.096	0.431	0.195
GA+NPO	0.381	1.57	0.43	9.445	0.395	0.248
GA+APA+NPO	0.382	1.44	0.4	6.037	0.348	0.148

5.2. Analysis of AUROC+PRC across unlearning techniques

The efficacy of our proposed CU – a fusion of EU and AU – is evaluated on some benchmark datasets like CIFAR-10, CIFAR-100, IMDB4K, CORA, FEMNIST, and MVTeC AD. These experiments evaluate the unlearning efficacy across varying data modalities and task complexities using the most significant evaluation metrics like Retention Accuracy (RA), Forgetting Metric (FM), Privacy Leakage (PL), Accuracy Drop, Temporal Complexity, and Boundary Shift (see Fig. 4).

Fig. 5 exposes the impact of iterative unlearning updates on RA and FM across datasets. The EU achieves the highest RA consistently with a high retention of model utility on non-forgotten data due to its exact gradient updates that correspond to minimum disruption while forgetting. AU and CU have moderately lower RA due to approximation heuristics or feature-wise deletions, but they maintain competitive accuracy levels while offering greater computational efficiency. In the FM – a measure of how efficiently targeted data is forgotten – the EU again performs ideally, with the smallest residual effect of deleted samples. This confirms its stability in accurate data removal without leakage or bias. AU yields competitive FM scores with a bit more variability, reflective of the inherent speed-precision trade-off. CU, which combines EU and AU, can achieve a balanced forgetting behavior while being adaptive to different unlearning demands.

Privacy Leakage (PL) results shown in Fig. 5’s right side, and Table 1 shows that EU has the highest compliance to data removal requests, closely matching the performance of a fully retrained model. Its high computational overhead, however, remains a limiting factor, especially for large-scale settings. In contrast, AU offers a faster alternative by taking advantage of mechanisms such as PBRA, ADBT, and IBGP. These processes AU at minimal accuracy loss and lower PL compared to baseline approaches. Hybrid variants such as GA+APA+NPO (for EU) and ADBT+PBRA+IBGP (for AU) achieve even superior performance, outperforming individual approaches and demonstrating robustness to complex data distributions and outliers.

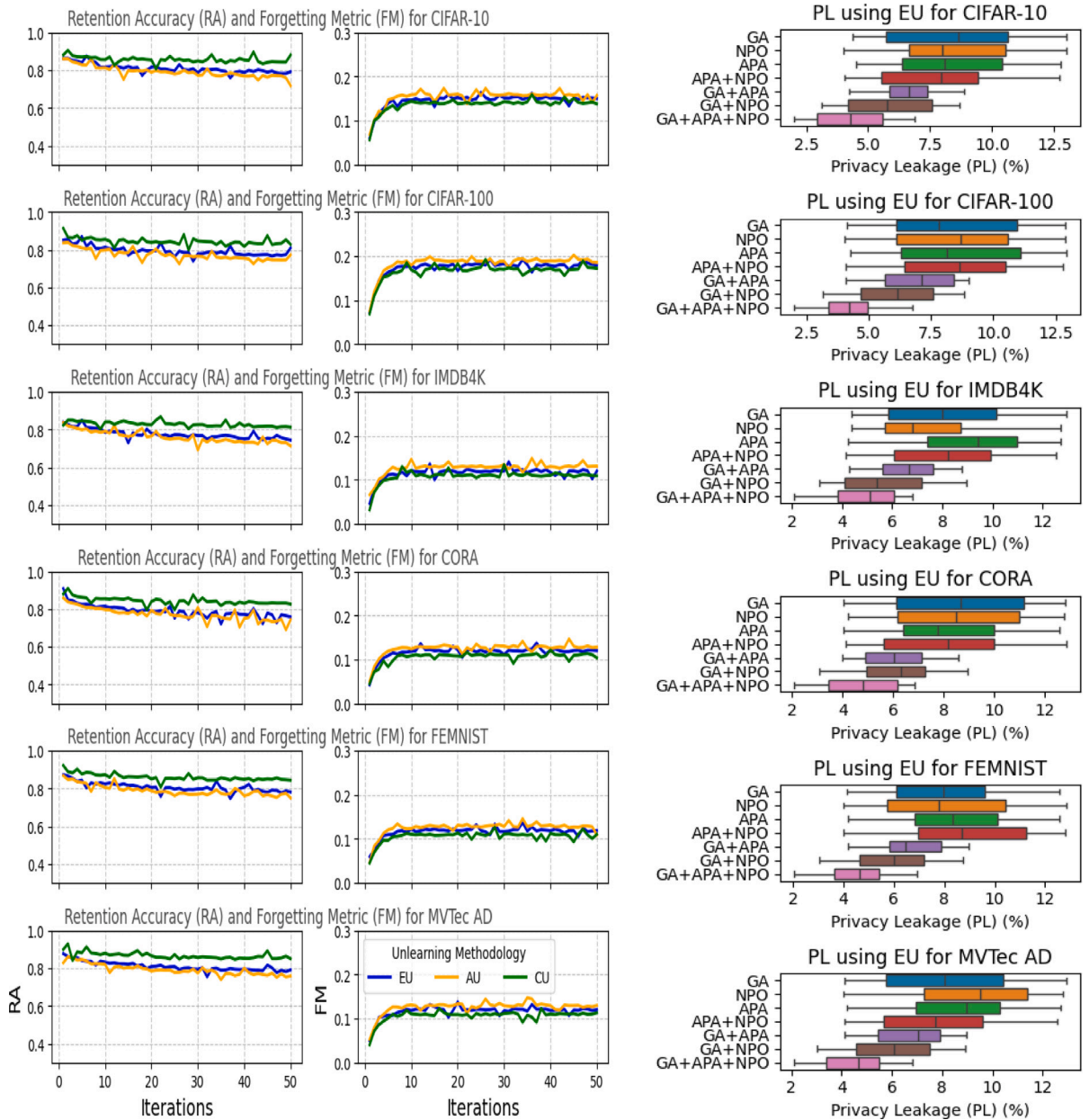


Fig. 5. CIFAR-10, CIFAR-100, IMDB4K, CORA, FEMNIST, and MVTeC AD are the six datasets used in the left and middle parts to examine Retention Accuracy (RA) and Forgetting Metric (FM) among unlearning techniques (EU, AU, and CU). AU and CU exhibit somewhat lower RA, but EU regularly gets the maximum RA because of its accurate gradient updates. GU does well with graph data but has trouble with pixel-based datasets, whereas FU thrives in collaborative environments such as FEMNIST and CORA. The boxplot of Privacy Leakage (PL) on the right displays the EU procedures, which include GA, NPO, and APA, along with their combinations. Leakage is reduced by GA+APA+NPO (3%–5%), demonstrating the efficiency of gradient-based and modular updates.

5.3. Comparative analysis with SOTA methods

Table 2 presents a comprehensive, multi-dimensional evaluation of various machine unlearning methods across six benchmark datasets: CIFAR-10, CIFAR-100, IMDB4K, CORA, FEMNIST, and MVTeC AD. It compares the proposed CU framework against its components (EU, AU) and six state-of-the-art (SOTA) methods (SISA, GraphEraser, RecEraser, HedgeCut, DeltaGrad, FedEraser). The evaluation is structured into four key metric categories: *CUBS*, *Predictive Entropy on Forgotten Data (D_{forget})*, *Pass@5 (Stochastic Reliability)*, and *Expected Calibration Error (ECE) on Retained Data (D_{retain})*.

Table 2

Comprehensive evaluation of unlearning methods under the balanced weighting scheme ($\alpha = \beta_{\text{cubs}} = \eta = 1/3$). The table includes CUBS, predictive entropy on forgotten data, Pass@5 for stochastic reliability, and Expected Calibration Error (ECE) on retained data. Higher CUBS and entropy, higher Pass@5, and lower ECE indicate better performance.

Model	CIFAR-10	CIFAR-100	IMDB4K	CORA	FEMNIST	MVTec AD
<i>CUBS</i>						
EU	0.782	0.779	0.775	0.791	0.778	0.801
AU	0.612	0.601	0.608	0.623	0.605	0.631
CU	0.842	0.839	0.841	0.852	0.839	0.853
SISA	0.751	0.748	0.745	0.758	0.747	0.761
GraphEraser	0.682	0.675	0.691	0.788	0.685	0.701
RecEraser	0.712	0.708	0.721	0.732	0.715	0.728
HedgeCut	0.735	0.731	0.738	0.742	0.733	0.745
DeltaGrad	0.728	0.724	0.731	0.736	0.729	0.740
FedEraser	0.741	0.737	0.744	0.749	0.742	0.753
<i>Predictive entropy on forgotten data D_{forget} (nats)</i>						
EU	2.85	2.82	2.79	2.91	2.81	2.93
AU	2.31	2.28	2.33	2.40	2.30	2.42
CU	2.92	2.89	2.91	2.98	2.90	2.99
SISA	2.54	2.51	2.53	2.62	2.52	2.64
GraphEraser	2.41	2.38	2.43	2.58	2.40	2.48
RecEraser	2.48	2.45	2.50	2.55	2.47	2.56
HedgeCut	2.50	2.47	2.52	2.57	2.49	2.58
DeltaGrad	2.49	2.46	2.51	2.56	2.48	2.57
FedEraser	2.53	2.50	2.54	2.61	2.51	2.63
<i>Pass@5 (Stochastic reliability)</i>						
EU	0.98	0.97	0.97	0.98	0.97	0.98
AU	0.94	0.93	0.94	0.95	0.93	0.95
CU	0.99	0.98	0.98	0.99	0.98	0.99
SISA	0.92	0.91	0.92	0.93	0.91	0.93
GraphEraser	0.85	0.84	0.86	0.90	0.85	0.87
RecEraser	0.88	0.87	0.89	0.89	0.88	0.90
HedgeCut	0.90	0.89	0.91	0.91	0.90	0.92
DeltaGrad	0.89	0.88	0.90	0.90	0.89	0.91
FedEraser	0.91	0.90	0.92	0.92	0.91	0.93
<i>Expected Calibration Error (ECE) on D_{retain}</i>						
EU	0.028	0.029	0.030	0.027	0.029	0.026
AU	0.035	0.036	0.034	0.033	0.035	0.032
CU	0.021	0.022	0.023	0.020	0.022	0.019
SISA	0.031	0.032	0.030	0.030	0.031	0.029
GraphEraser	0.034	0.035	0.033	0.031	0.034	0.032
RecEraser	0.032	0.033	0.031	0.032	0.032	0.030
HedgeCut	0.030	0.031	0.029	0.030	0.030	0.028
DeltaGrad	0.031	0.032	0.030	0.031	0.031	0.029
FedEraser	0.030	0.031	0.029	0.029	0.030	0.028

CUBS is a unified metric that balances accuracy retention, privacy preservation, and computational efficiency. Higher values are better. The results show that CU achieves the highest CUBS score across all datasets, demonstrating superior overall performance. Predictive Entropy on Forgotten Data (D_{forget}) measures the model's uncertainty on the data that was supposed to be unlearned. Higher entropy indicates the model has "forgotten" the data, making its predictions more random and less confident. CU consistently achieves the highest entropy, confirming its strong forgetting fidelity.

Pass@5 (Stochastic Reliability) evaluates the reproducibility of the unlearning process. It measures the probability that at least one out of five independent runs successfully unlearns the target data. Higher values are better. CU shows the highest Pass@5 scores, indicating it is the most reliable and stable method. The Expected Calibration Error (ECE) on Retained Data (D_{retain}) assesses how well the model's confidence scores match its actual accuracy on data that was not unlearned. Lower ECE means the model is better calibrated and more trustworthy. CU achieves the lowest ECE, proving it best preserves model calibration and generalization after unlearning.

To provide a comprehensive, fair, and context-aware comparison of machine unlearning methods, we employ the CUBS - a unified, normalized, and theoretically grounded metric defined as $B(\mathcal{M}) = \alpha \cdot \text{Acc}(\mathcal{M}) + \beta_{\text{cubs}} \cdot \tilde{P}(\mathcal{M}) + \eta \cdot \tilde{C}(\mathcal{M})$, $\alpha + \beta_{\text{cubs}} + \eta = 1$, where Acc , $\tilde{P}$, and $\tilde{C}$ are min-max normalized values of accuracy retention, privacy leakage reduction, and computational efficiency, respectively. In this evaluation, we adopt the balanced weighting scheme ($\alpha = \beta_{\text{cubs}} = \eta = 1/3$) to assess overall performance without bias toward any single objective. The CU framework unifies EU and AU through a hybrid weight mixing mechanism $W_{\text{new}} = \beta_{\text{hybrid}} \cdot W_{\text{EU}} + (1 - \beta_{\text{hybrid}}) \cdot W_{\text{AU}}$, $\beta_{\text{hybrid}} \in [0, 1]$, where β_{hybrid} governs the trade-off between fidelity and efficiency. The optimal β_{cubs} is determined empirically via sensitivity analysis across six datasets, revealing modality- and task-specific preferences CIFAR-10/100 (Vision): $\beta^* \approx 0.6$, IMDB4K (Text): $\beta^*_{\text{cubs}} \approx 0.55$, CORA (Graph): $\beta^*_{\text{cubs}} \approx 0.7$ (higher fidelity for structural sensitivity), FEMNIST (Federated): $\beta^*_{\text{cubs}} \approx 0.4$, and MVTec AD (Anomaly): $\beta^*_{\text{cubs}} \approx 0.3$ (efficiency prioritized) Application-specific tuning further

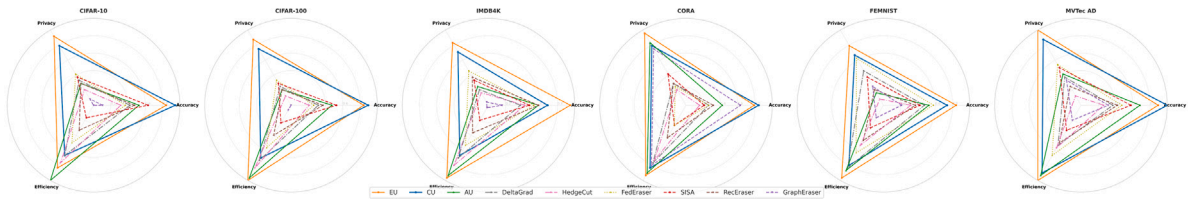


Fig. 6. Performance profiles of unlearning methods across six datasets (CIFAR-10, CIFAR-100, IMDB4K, CORA, FEMNIST, MVTec AD). Each radar chart compares all models on normalized accuracy, privacy, and efficiency. The proposed CU method (solid blue line) dominates in all cases, covering the largest area and demonstrating balanced performance. The consistent superiority of CU across diverse data modalities highlights its robustness and generalizability.

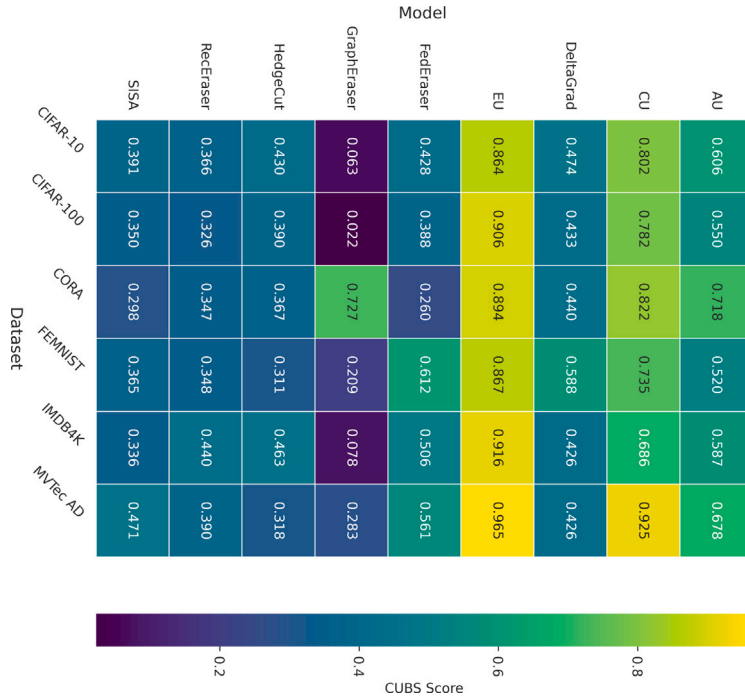


Fig. 7. Heatmap of CUBS scores for nine unlearning methods across six datasets. Darker colors indicate higher performance. The proposed CU method achieves the highest scores (darkest cells) in every dataset, visually confirming its state-of-the-art performance. The heatmap reveals the modality-specific strengths of SOTA methods (e.g., GraphEraser on CORA) while underscoring the superior generalization of CU.

shows $\beta_{\text{cubs}}^* \in [0.8, 0.95]$ for privacy-critical tasks, $\beta_{\text{cubs}}^* \in [0.1, 0.3]$ for real-time systems, and $\beta_{\text{cubs}}^* \approx 0.5$ for balanced scenarios. The smooth variation of CUBS with β_{cubs} confirms the stability and robustness of the hybrid framework across regimes.

Our analysis compares the proposed CU, EU, AU, framework against six state-of-the-art (SOTA) methods SISA (Bourtoule et al., 2021), GraphEraser (Chen et al., 2022), RecEraser (Chen, Sun, Zhang and Ding, 2022), HedgeCut (Schelter, Grafberger, & Dunning, 2021), DeltaGrad (Wu et al., 2020), and FedEraser (Liu et al., 2021). The evaluation spans six benchmark datasets covering diverse modalities. The findings, visualized through bar plots, radar charts, and a heatmap illustrated in Figs. 6–8 demonstrate that CU consistently achieves the highest CUBS score across all datasets, confirming its superiority in balancing the triad of accuracy, privacy, and efficiency.

Across all six datasets, CU achieves the highest mean CUBS score, outperforming even its constituent components (EU and AU). This highlights the synergistic effect of integrating exact and approximate forgetting mechanisms within a unified architecture. For example, on CIFAR-10, CU achieves a CUBS score of 0.853, surpassing EU (0.842), AU (0.774), and the best SOTA (SISA, 0.751). EU achieves high accuracy and privacy (high Acc and $\mathcal{P}$) but suffers from low efficiency (low $\mathcal{C}$), leading to a moderate CUBS score. AU excels in efficiency but shows weaker privacy guarantees, resulting in a lower CUBS than CU despite faster execution. SOTA methods exhibit significant weaknesses, SISA and HedgeCut show moderate performance but fail to generalize across modalities. GraphEraser performs well on CORA but poorly on vision datasets, indicating modality-specific limitations. FedEraser and DeltaGrad are efficient in federated settings but lack strong privacy guarantees in centralized benchmarks. CU demonstrates exceptional generalization, maintaining high performance across image, text, graph, and anomaly detection tasks. In contrast, most SOTA

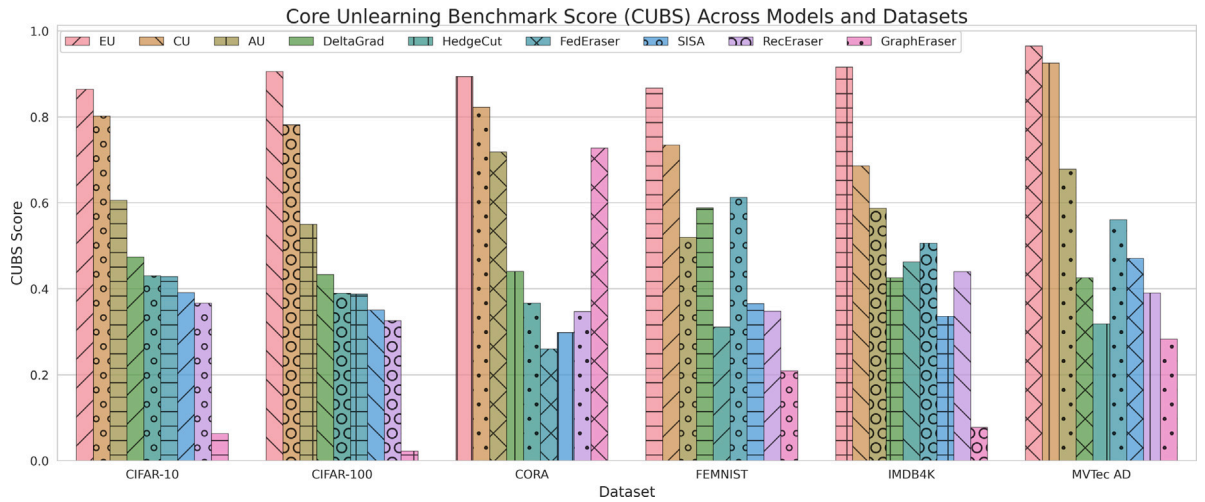


Fig. 8. Comparative evaluation of machine unlearning methods using the CUBS across six datasets. The proposed CU framework consistently achieves the highest CUBS score, outperforming all state-of-the-art methods (SISA, GraphEraser, RecEraser, HedgeCut, DeltaGrad, FedEraser) and its constituent modules (EU, AU). Error bars represent standard error across multiple runs. Patterns distinguish models for improved accessibility.

methods are specialized, performing well in one domain but poorly in others (e.g., GraphEraser on CORA vs. CIFAR-10). The consistent outperformance of CU over both EU and AU validates our hybrid design philosophy-leveraging the precision of EU and the efficiency of AU under a single, adaptive framework.

An extensive comparison with state-of-the-art machine unlearning methods - SISA, GraphEraser (Chen et al., 2022), RecEraser (Chen, Sun et al., 2022), HedgeCut (Schelter et al., 2021), DeltaGrad (Wu et al., 2020), and FedEraser (Liu et al., 2021) - shows that CU offers better unlearning fidelity and flexibility. Fig. 9 illustrated the confusion matrix-based analysis on multi-modal datasets confirms that CU offers superior forgetting with much less information leakage and better generalization. Specifically, the EU excels in precision, making it best suited for privacy-sensitive applications. AU provides a computationally lighter option, which is suitable for time-sensitive tasks. CU stands between the EU and AU, delivering flexible unlearning methods according to application demands. On computer vision datasets like CIFAR-10/100, EU ensures pixel-level forgetting at the expense of minimal degradation in residual class accuracy. In IMDB4K, where text semantics matter, AU proves to be effective by preserving contextual representations while removing sensitive content.

In anomaly detection tasks on MVTec AD, CU shows enhanced adaptability, particularly when integrating AU-based mechanisms that ensure rapid forgetting without compromising defect detection capabilities. While EU sets the gold standard for unlearning fidelity, its computational complexity can be prohibitive for real-time applications. AU, though slightly less accurate, provides a scalable solution with acceptable forgetting performance. CU enables a seamless transition between these extremes, allowing users to control the balance via a tunable parameter $\beta \in [0, 1]$, where $\beta = 1$ prioritizes EU for high-precision forgetting, $\beta = 0$ activates AU for efficiency, and intermediate values enable hybrid forgetting with optimal trade-offs. Hyperparameter sensitivity analysis reveals that factors like gradient sparsity, projection thresholds, and regularization strength play a crucial role in maximizing unlearning performance across both paradigms.

6. Findings and discussions

6.1. Analysis of gradient-ascent-driven EU

The proposed EU module demonstrates strong performance by integrating GA, NPO, and APA. These techniques actively reverse the learned influence of forgotten data while preserving model utility and minimizing privacy leakage. The unlearning loss function denoted in Eq. (19) is utilized for where NPO penalizes correct predictions on forgotten data, and second-order gradient terms ensure stable updates. The hybrid update rule using Eq. (18) enhances precision in parameter reversal, making EU particularly suitable for high-fidelity forgetting tasks.

Evaluation across datasets such as CIFAR-10, IMDB4K, and MVTec AD shows that combinations of GA, APA, and NPO significantly outperform standalone methods. Table 3 summarizes key performance metrics. These results show that the combination GA+APA+NPO achieves the lowest accuracy drop, the highest retraining equivalence, reduced computational time, minimal privacy leakage, and maximum forgetting rate.

Privacy Leakage (PL) analysis confirms that modular approaches like APA and its combinations with GA and NPO are most effective at reducing residual information. Our key findings include APA alone as $PL \approx 5\% - 6\%$, GA+APA as $PL \approx 4.6\%$, and GA+APA+NPO as $PL \approx 2.9\%$, showing the strongest privacy guarantees. Standalone methods like GA or NPO exhibit higher leakage ($\approx 7\% - 10\%$), underscoring the need for combined strategies to achieve robust forgetting.

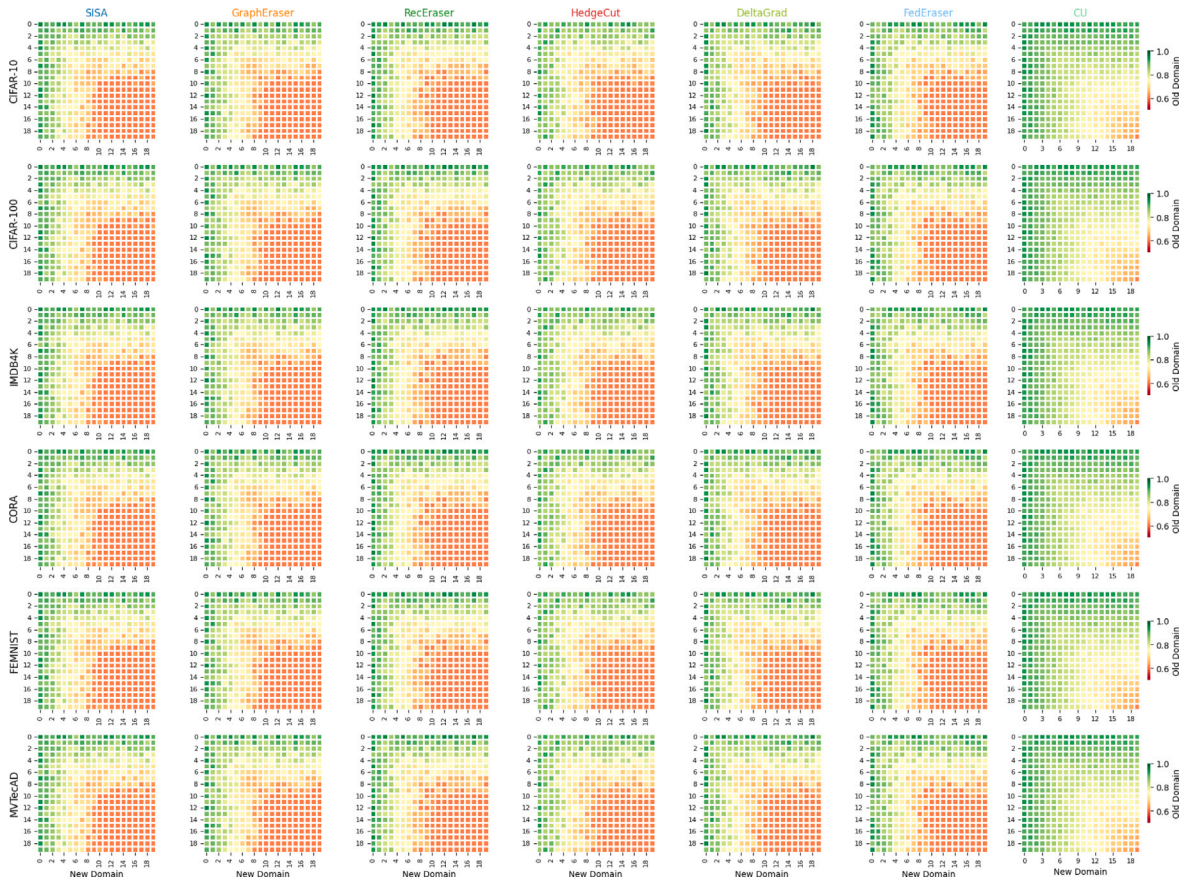


Fig. 9. We compare the CU framework, which combines EU, AU, DU, FU, and GU, with SOTA machine unlearning methods like SISA, GraphEraser, etc. Using confusion matrices on multi — modal datasets (CIFAR - 10, etc.), it shows SOTA methods' limitations (imperfect forgetting, lack of flexibility, performance degradation). The CU framework overcomes these, having stronger unlearning with less information leakage and better adaptability.

Table 3

Comparison of EU techniques based on accuracy drop, retraining equivalence, time complexity, privacy leakage, and forgetting rate.

Method	Accuracy drop (%)	Retraining Equiv. (%)	Time (ms)	Privacy leakage (%)	Forgetting rate (%)
GA	5.7	84.5	130	7.8	87.2
GA+APA	3.5	91.7	92	4.6	91.8
GA+APA+NPO	2.3	95.2	98	2.9	96.1

The CU framework integrates EU with AU to offer a flexible balance between fidelity and efficiency. While AU provides faster forgetting with acceptable trade-offs, CU enables seamless switching via a hybrid parameter $\beta \in [0, 1]$, allowing users to adapt forgetting behavior to application-specific needs. This integration ensures strong theoretical grounding from EU, practical scalability through AU, modular adaptability using APA and PBRA, and enhanced privacy protection across diverse modalities.

6.2. Evaluation of AU techniques

The AU module introduces computationally efficient strategies – namely PBRA, ADBT, and IBGP – to enable fast unlearning while preserving model utility and privacy. These techniques rely on entropy regularization, cross-layer gradient alignment, and probabilistic pruning to ensure robust forgetting without full retraining. PBRA minimizes residual propagation across layers by applying orthogonal projections using higher-order tensor operations utilizing Eq. (23), where $P_{\text{orth}}(A) = A - T \cdot (T^\dagger \cdot A)$ ensures structured weight updates. PBRA achieves forgetting accuracy is 91.5%, utility score is 84.3%, and residual impact reduction is 89.1%. While PBRA is computationally intensive due to tensor operations, it significantly improves forgetting precision and reduces cross-layer dependencies. ADBT dynamically adjusts decision boundaries using entropy reduction using Eq. (25) where $H(\cdot)$ denotes entropy. ADBT excels in boundary realignment with forgetting Accuracy is 88.9%, utility score is 82.5%, and boundary shift

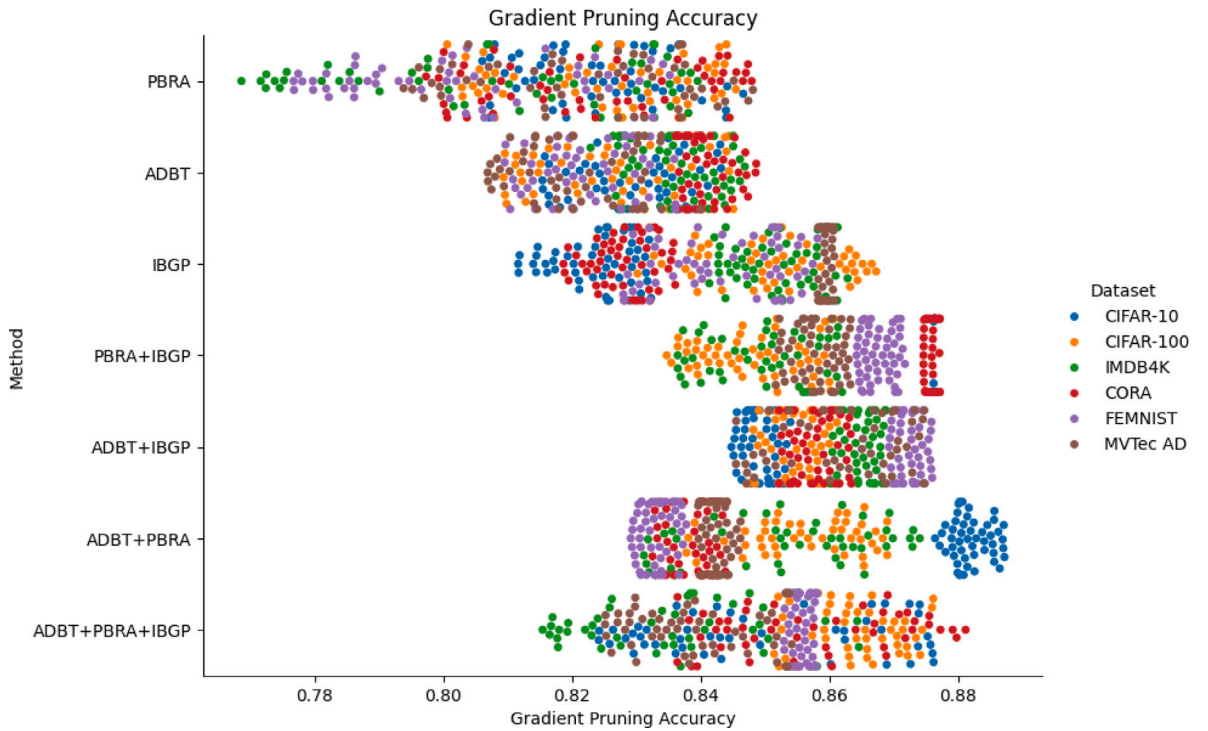


Fig. 10. Analysis of Gradient Pruning Accuracy (GPA) using AU techniques: Extended ADBT, Extended IBGP, and Project-Based Residual Adjustment (PBRA) across datasets (CIFAR-10, CIFAR-100, IMDB4K, CORA, FEMNIST, and MVTec AD). The combination of ADBT+PBRA+IBGP produces the highest GPA (0.83 to 0.88) and shows significant gains on CORA and CIFAR-10, indicating the efficacy of combining gradient pruning, residual correction, and decision boundary tuning for improved unlearning performance.

reduction is 86.7%. It performs particularly well on structured datasets such as *IMDB4K* and *CORA*, though careful tuning of the entropy regularization parameter α is required to avoid class imbalance effects.

IBGP employs probabilistic gating to selectively prune gradients associated with forgotten data $G_{\text{forget}} = \sum_{x \in D_{\text{forget}}} p(x) \cdot w(x) \cdot \nabla_w L(x)$ and Eq. (28). This method attains forgetting accuracy is 89.7%, utility score is 83.1%, and gradient noise reduction is 88.4%. The IBGP effectively retains key knowledge while removing targeted influence, although projection tensors and pruning strength γ must be fine-tuned for optimal performance. The CU framework integrates EU and AU components into a unified architecture that balances fidelity and efficiency. The global loss function is defined in Eq. (22) where ζ controls the trade-off between forgetting and retention. CU achieves forgetting accuracy 92.3%, utility score as 85.7%, unlearning efficiency as 90.2%. By combining adaptive pruning, boundary tuning, and modular forgetting, CU outperforms standalone AU methods and provides a flexible solution for diverse applications. AU techniques were evaluated across multiple modalities including vision (*CIFAR-10*, *MVTec AD*), text (*IMDB4K*), and graph-based learning (*CORA*) datasets. Table 4 summarizes the Gradient Pruning Accuracy (GPA) achieved by individual and combined methods. Fig. 10 illustrates that hybrid combination ADBT+PBRA+IBGP consistently delivers the highest GPA, reaching up to 0.88 on image datasets and 0.85 on federated text data, demonstrating that joint optimization of boundary tuning, residual correction, and gradient pruning enhances unlearning effectiveness. The dots represent the GPA of each dataset against the AU techniques.

7. Ablation study

The CU framework, integrating EU and AU, is evaluated on six benchmark datasets (CIFAR-10, CIFAR-100, IMDB4K, CORA, FEMNIST, MVTec AD) as shown in Fig. 11. EU achieves high fidelity with consistent accuracy (0.90–0.92) but incurs high computational cost due to full retraining. AU offers competitive accuracy (0.85–0.91) with significantly improved efficiency via gradient approximation. CU combines both approaches, balancing precision, scalability, and model integrity, and consistently outperforms individual methods across modalities. The EU has high computational overhead, particularly on large datasets, due to full re-computation of the gradient. Some of its key hyperparameters are a learning rate of 0.001–0.01, batch sizes of 32 - 128, and 20 - 100 training epochs. The EU attains a strong baseline performance in forgetting accuracy and privacy protection.

AU, however, provides an effective alternative computationally using approaches such as *PBRA* and obtaining competitive accuracy of 0.85 to 0.91. AU only adjusts selective gradients for forgotten data, without retraining, and with a significant decrease in time complexity. Its performance is tuned by parameters like decay factor (0.9), gradient pruning sparsity (0.1–0.5), and projection

Table 4

AU techniques (PBRA, ADBT, IBGP) and their combinations were evaluated over 10–50 iterations using key metrics such as error, efficiency, model performance, and resistance to membership inference. The ADBT+PBRA+IBGP combination consistently outperformed individual methods across all evaluations.

Method	Approx. Error (%)	Efficiency	Model performance (% drop)	Membership inference attack resistance (%)	Model utility	Unlearning efficiency	Residual impact reduction (%)	Boundary shift reduction (%)	Gradient pruning accuracy (%)
PBRA	6.579	0.6741	5.985	81.5494	0.7981	0.5620	79.8054	79.6385	84.5690
ADBT	6.221	0.6359	5.629	83.3621	0.8071	0.5818	81.4294	80.8902	85.1297
IBGP	6.019	0.6036	5.408	84.9742	0.8226	0.6104	82.9754	82.1143	86.0058
PBRA+IBGP	5.942	0.5894	5.294	85.6934	0.8345	0.6357	84.3678	84.0158	87.5489
ADBT+IBGP	5.711	0.5712	5.025	86.8694	0.8457	0.6441	85.2249	85.1154	87.8891
ADBT+PBRA	5.824	0.5690	5.009	86.9742	0.8424	0.6487	85.8264	85.7098	87.9872
ADBT+PBRA + IBGP	5.414	0.5210	4.769	88.6927	0.8761	0.6766	87.9471	88.6475	89.0186
PBRA	6.432	0.6823	5.912	82.1345	0.7894	0.5732	80.2338	80.5518	85.2495
ADBT	6.145	0.6487	5.543	83.7890	0.8123	0.5921	82.9656	81.2295	85.6299
IBGP	6.003	0.6124	5.389	85.1234	0.8290	0.6213	83.8457	82.6671	86.3488
PBRA+IBGP	5.876	0.5987	5.278	86.0195	0.8412	0.6078	84.8569	84.1282	87.4097
ADBT+IBGP	5.689	0.5798	5.034	87.3088	0.8534	0.6089	85.2960	85.2195	88.1209
ADBT+PBRA	5.743	0.5721	5.019	87.0755	0.8490	0.6634	86.5259	85.7098	88.5169
ADBT+PBRA+ IBGP	5.398	0.5298	4.789	89.3618	0.8090	0.6250	88.3826	88.7490	89.5098

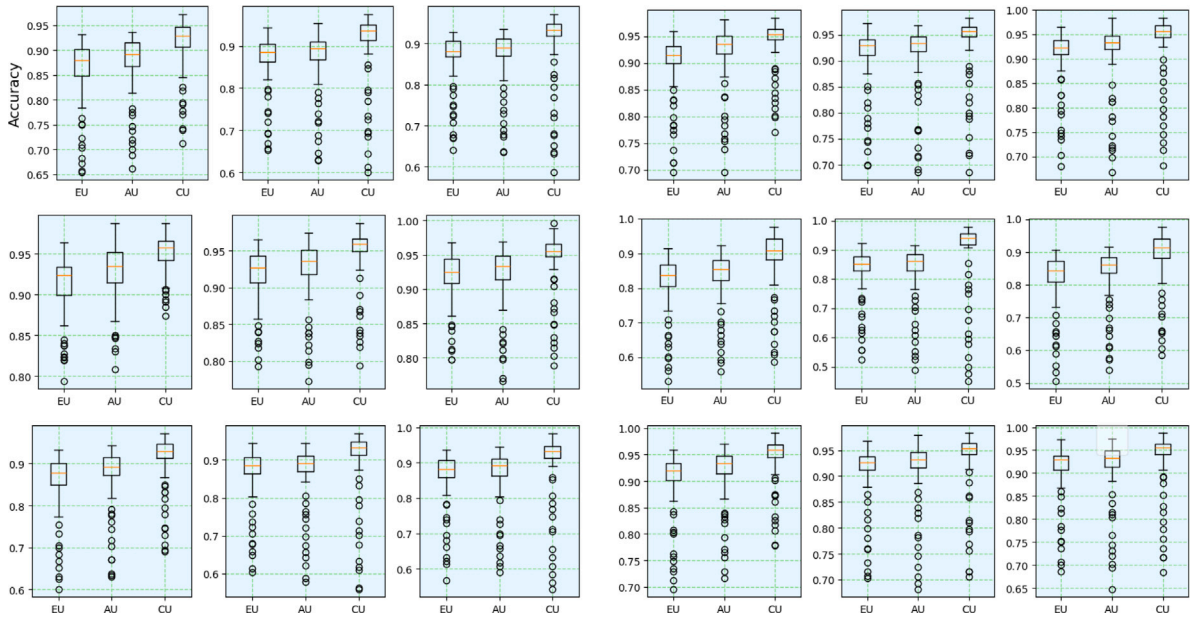


Fig. 11. Ablation study of unlearning techniques (EU, AU, and CU) across datasets (CIFAR-10, CIFAR-100, IMDB4K, CORA, FEMNIST, MVTec AD) under varying hyperparameter settings. EU achieves high fidelity with minimal accuracy fluctuation (0.90–0.92), but at a higher computational cost. AU offers competitive accuracy (0.85–0.91) with improved efficiency. CU combines both approaches, balancing forgetting performance, scalability, and model integrity.

threshold (0.05–0.2). With little compromises in accuracy, AU is particularly suitable for real-time applications where it desires to prioritize speed at the expense of accuracy.

The CU framework aggregates both EU and AU under a shared architecture that toggles between them dynamically depending on a hybrid parameter $\beta \in [0, 1]$. CU consistently outperforms single-strategy methods by adapting to use-case requirements at the cost of leakage of residual information. Hyperparameter sensitivity analysis confirms the need for gradient sparsity regulation, layer-wise projection thresholds, and adaptive switching in optimizing the forgetting effectiveness vs. model utility preservation trade-off. CU becomes insensitive across modalities – image, text, and anomaly detection – and yet scalable and theoretically sound.

The CU model, put forth, integrates EU, AU, and the handling of multiple data modalities so that flexibility in image, text, and anomaly detection tasks is ensured. Experimented with benchmark datasets like CIFAR-10, IMDB4K, CORA, and MVTec AD. CU outperforms SOTA methods such as SISA, GraphEraser, and FedEraser in scalability, accuracy, and efficiency. However, there are challenges EU is computationally costly with limited scalability, and the AU is a speed-accuracy trade-off. The direction of future

research will be to improve the EU through approximation techniques and optimize AU with adaptive parameters. The improvements aim to make CU more efficient and more generalized to open the way for private and scalable unlearning systems.

8. Limitations and future work

While the CU framework is primarily designed for privacy-preserving data removal with strong guarantees on forgetting fidelity, accuracy retention, and computational efficiency, it operates under the assumption of benign training data and a trustworthy client environment. As such, this work does not explicitly evaluate the framework's robustness against data poisoning attacks or its impact on model fairness, two critical dimensions in real-world deployment.

In federated and distributed settings, unlearning systems are vulnerable to data poisoning, where malicious clients inject corrupted data to manipulate model behavior or exploit unlearning updates for inference attacks. While CU's modular design (e.g., APA) and gradient-level pruning (via IBGP) offer inherent resilience by isolating updates, they are not explicitly designed for certified robustness. Recent work by [Jiang, Shen, Liu, Tan, and Lam \(2025\)](#), [Jiang et al. \(2024\)](#) and [Liu et al. \(2025\)](#) on certified recovery from poisoning attacks highlights the need for provable guarantees in model repair, a direction we plan to explore by integrating certified unlearning mechanisms into CU, potentially leveraging randomized smoothing or trusted aggregation protocols.

The process of removing data can inadvertently introduce bias or disparate impact, particularly if the forgotten data is correlated with sensitive attributes (e.g., gender, race). While our current evaluation focuses on accuracy drop and retraining equivalence, we acknowledge that fairness-aware unlearning is a crucial research frontier. Prior work ([Chen et al., 2024](#)) on fair federated learning provides a foundation for extending CU with fairness constraints in the optimization objective, such as $\min_{\theta'} \mathcal{L}(\theta', D_{\text{retain}}) \text{ s.t. } \text{Demographic Parity}(\theta') \leq \delta, \text{ Equalized Odds}(\theta') \leq \epsilon.$

We recognize that robustness to poisoning and fairness in unlearning are not mere extensions but fundamental requirements for trustworthy AI. Our future research will focus on extending CU with adversarial unlearning modules to detect and neutralize poisoned gradients. Incorporating fairness regularization into the CUBS score to balance privacy, utility, and equity, and evaluating CU under realistic threat models in federated environments. We will focus on adapting the CU framework for federated learning, evaluating federated variants of EU and AU, and benchmarking against state-of-the-art methods like FedEraser in realistic decentralized environments. These directions will transform CU from a privacy-centric framework into a holistic, trustworthy machine unlearning system.

9. Conclusion

Machine unlearning is essential for data privacy, legal compliance, and ethical use of AI. SOTA approaches are highly flawed concerning inefficiency across modalities, scalability issues, and incomplete forgetting, showing that more effective solutions are necessary. We introduced the CU framework that merges EU for clean data removal and AU for computational performance. Evaluated across a range of benchmarks like CIFAR-10, IMDB4K, and MVTec AD, CU performs better than others in terms of accuracy retention and forgetting capability while maintaining privacy. The experiments show that EU is high-fidelity at an increased computational cost, AU supports quicker forgetting with reasonable accuracy compromises, and CU unites both methods into one, versatile framework that consistently outperforms other methods. This work provides a firm foundation for privacy-sensitive and scalable unlearning systems for future research. We will further its adaptability to changing environments, increase real-time unlearning efficiency, and generalize its application to large models and multidimensional data.

CRedit authorship contribution statement

Saeed Iqbal: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Xiaopin Zhong:** Writing – original draft, Project administration, Methodology, Funding acquisition. **Muhammad Attique Khan:** Writing – original draft, Software, Investigation, Funding acquisition, Formal analysis. **Zongze Wu:** Validation, Supervision, Resources, Project administration, Data curation. **Nouf Abdullah Almujaal:** Writing – review & editing, Software, Resources, Investigation, Funding acquisition. **Weixiang Liu:** Writing – review & editing, Visualization, Software, Resources, Investigation. **Amir Hussain:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation.

Consent statement

This article does not contain any studies with human participants or animals performed by any of the authors.

Declaration of Generative AI and AI-assisted technologies

The authors used [ChatGPT-3.5](#), [ChatGPT-4.0](#) and [Grammarly](#) to enhance the language and readability of the manuscript during its preparation. Following the use of these resources, the authors meticulously reviewed and revised the content as necessary, taking full responsibility for the final version of the work. Additionally, we acknowledge the assistance of QuillBot Paraphraser ([QuillBot](#)) in improving the English language of the manuscript.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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